

Toward User-Centered Multimodal Conversational Recommendation:

An Information-Needs Perspective

by

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Abstract

Multimodal conversational recommender systems (CRS) are becoming increasingly important as natural language interfaces spread and recommendation moves into richer, more complex settings. However, we find that while Multimodal Large Language Model (MLLM) based systems can generate locally plausible responses, informational exchange during recommendation is poorly supported. Despite this, multimodal CRSs are still evaluated mainly through recommendation and response quality. We therefore argue that limited formalization of Information-Needs in multimodal CRS constrains both evaluation and system design.

To address this, we introduce a two-part study and framework. First, we create and analyze VOGUE, a human–human multimodal conversation dataset grounded in the fashion domain. Our analysis highlights the importance of both eliciting preferences and also managing information exchange between user and recommender. We therefore propose an Information-Needs formalization for multimodal CRS, with proactive versus retroactive needs and user- versus expert-oriented needs. This formalization motivates richer evaluation of success in CRS, and enables deeper analysis of failure modes in MLLM CRS systems.

Second, we introduce a test bench for studies of MLLM-driven CRS. Using this platform, we find that frontier MLLM baselines exhibit persistent weaknesses in multi-turn strategy, use of implicit signals, and decision support, despite local coherence. Motivated by our findings, we introduce and analyze several design variations and demonstrate uplift when we explicitly model for Information-Needs and multi-turn planning.

These findings motivate a move toward multimodal CRS architectures that explicitly represent Information-Needs and thus ground recommendation in decision-relevant exchange.

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Chapter 1

Introduction

Conversational recommender systems (CRS) extend recommendation from static ranking to multi-turn interaction. Rather than producing ranked lists or item ratings, a CRS engages the user in dialogue to elicit preferences, refine constraints, narrow the candidate set, and support selection [1, 2]. In multimodal settings, that interaction is additionally grounded in other modalities, such as images, reviews, or testimony. These settings better reflect realistic decision problems, but they also expose a limitation in how conversational recommendation is commonly formulated and evaluated.

Most CRS work treats conversation primarily as a mechanism for improving preference estimation and item ranking [1, 2]. In this literature, user modeling is often reduced to preference modeling, and system quality is assessed mainly through recommendation or response quality. That framing has been useful, but it is incomplete for realistic multimodal decision settings. Users often need more than a system that identifies items consistent with stated preferences. They also need help refining partially formed criteria and judging plausible alternatives, especially when important properties are not always user-available [3, 4, 5].

Conversational recommendation is fundamentally a decision support problem. Users bring private goals and contextual constraints, but often cannot fully articulate their internal motivations and knowledge. The system, by contrast, has access to catalog structure and comparative evidence that the user does not [6]. A successful interaction must bridge this informational asymmetry. In practice, users do not always state their intents, motivations, or needs perfectly. A system may therefore align well with expressed preferences while still failing to provide support for the user’s true need. The problem is especially acute in multimodal CRS, where rich item representations may seem informative without resolving the user’s uncertainty [5]. Even when the interaction remains locally coherent, poor support can suggest a level of decision confidence that the conversation

has not earned.

We therefore treat multimodal CRS as a problem of information exchange, in addition to preference inference. We use the term *Information-Needs* (INs) broadly to refer to the information exchanges necessary in the decision-making process. INs include not only explicit questions, but also partly expressed preferences, unresolved tradeoffs, and other decision-relevant considerations from either party. Preference elicitation is therefore not separate from addressing INs, but one aspect of it. Related concerns have been studied in conversational information seeking, where systems are designed to manage evolving needs and uncertainty [7, 6, 8]; however, they remain underemphasized in multimodal CRS, where reactive preference clarification still dominates modeling and evaluation [1].

This thesis addresses the Information-Need gap in two parts. First, we develop and analyze VOGUE, a human-human multimodal conversational recommendation dataset grounded in fashion. VOGUE pairs dialogues with item-level ratings and satisfaction signals, enabling analysis of conversational success beyond final recommendation alignment alone. Our analyses show that strong preference alignment does not fully characterize a successful interaction, and that good multimodal recommendation often depends on whether the recommender helps address decision-relevant uncertainties and informational asymmetries. These observations motivate a formalization of INs in multimodal CRS that distinguishes between user- and expert-oriented needs, and between needs already made salient in interaction (Retroactive) and those that must be anticipated (Proactive).

Second, we introduce an experimental test bed for live studies of MLLM-driven multimodal CRS. In this end-to-end setting, strong multimodal baselines exhibit persistent weaknesses in implicit-feedback understanding, preference and intent clarification, and coherent multi-turn interaction. Rather than progressively refining their understanding of the user, they often move too quickly into recommendation, rely on broad candidate shotgunning, and use explanation only weakly as a tool for comparison and uncertainty reduction. These results complement the VOGUE analyses by showing that multimodal conversational recommendation still involves challenges beyond IN satisfaction, including reasoning over recommendation sets, using indirect user evidence, and sustaining a staged interaction strategy across turns. Taken together, these findings motivate more structured system support, which we illustrate through designs that use explicit interaction state as a mechanism for supporting interaction planning and grounded explanation.

We argue that current multimodal CRS evaluation is too narrow because it overweighs preference alignment and poorly emphasizes IN satisfaction. We develop an IN-oriented perspective on this problem through human-human multimodal dialogue in VOGUE, then uses live MLLM-based experiments to identify additional interaction failures that motivate

richer state and planning support.

The contributions of this thesis are as follows:

- C1. Information-Needs formalization.** We introduce an IN-oriented formalization of multimodal conversational recommendation that treats preference elicitation as one part of a broader decision-relevant information exchange needed for supported choice. We represent INs as question–answer requirements and organize them by source and status.
- C2. Vogue.** We introduce and analyze VOGUE, a human–human multimodal conversational recommendation dataset that supports joint study of preference alignment, IN satisfaction, and user satisfaction. Using VOGUE, we show the limits of evaluating conversational success without considering informational support.
- C3. Empirical analysis of MLLM-based multimodal CRS.** We present an experimental test bed for end-to-end multimodal conversational recommendation and use it to characterize recurring weaknesses of current MLLM-based baselines, including poor use of implicit feedback, limited preference and intent clarification, and weak multi-turn interaction management.
- C4. Design implications.** Drawing on both the VOGUE analysis and the live interaction results, we derive design implications for multimodal CRS, especially the need for richer interaction state, staged planning, and more grounded comparative explanation. We illustrate these implications through example state-based designs.

Chapter 2

Foundations and Related Literature

2.1 A Brief Taxonomy of Multimodal CRS

CRS differ from task-oriented dialogue systems in that they are typically not organized around completing a predefined external action through a fixed schema. Instead, conversational recommendation operates in an open-ended preference space where user goals may be partial or evolving, and relevant criteria may emerge during interaction [4]. Most CRS therefore share a similar core structure: they interpret user input, maintain conversational state, retrieve or rank candidate items, and choose the next conversational move [1].

Existing taxonomies distinguish CRS by how they represent preferences, structure interaction, use supervision, and generate recommendations [1, 2]. These distinctions separate, for example, systems that rely on symbolic or slot-based user models from those using latent neural representations. They also distinguish systems that structure dialogue around direct elicitation, critiquing, mixed initiative, or free-form generation. Some pipelines keep recommendation and response generation modular, while newer architectures increasingly combine user understanding, ranking, and response generation within a single model.

Multimodal CRS inherit these distinctions while adding a further challenge: interaction is grounded not only in language, but also in item-associated modalities such as images [1, 2]. The system must therefore reason both about user preferences and about the role of visual and textual evidence during interaction. More recent LLM- and MLLM-based approaches further blur earlier boundaries between user modeling, recommendation, and dialogue management by performing multiple sub-tasks within a single interface [9, 10]. As a result, many CRS taxonomies still distinguish methods by how they acquire and use preference signals, while giving less explicit attention to how interaction helps users compare options and resolve uncertainty [1, 2].

2.2 Conversational Information Seeking

Conversational information seeking (CIS) provides a useful adjacent perspective. In CIS, dialogue is framed not just as a means of expressing a goal, but as a process of identifying and satisfying an evolving IN [7]. Interaction therefore centers on clarification, uncertainty management, context accumulation, and the retrieval or presentation of evidence relevant to the user’s underlying need. Foundational formulations of conversational search accordingly emphasize mixed-initiative behavior, clarification, and context-sensitive adaptation as central system capabilities [6].

This perspective makes the role of missing information explicit. A conversational agent may ask follow-up questions, reformulate a request, retrieve supporting content, or adapt its behavior based on what has already been established in the interaction. Benchmarks such as CAsT reflect this view by evaluating systems on their ability to retrieve and rank relevant content under conversational context, rather than only on whether they eventually arrive at a successful outcome [8]. In CIS, evidence acquisition and uncertainty reduction are treated more as part of the task itself than as behavior that emerges incidentally during interaction.

CIS therefore treats evolving needs, clarification, and evidence acquisition as central interaction goals, in contrast to the stronger focus on preference and recommendation quality in CRS.

2.3 Preference- and Constraint-Centric Conversational Recommendation

A large portion of the CRS literature treats interaction primarily as a way to elicit, refine, or infer user preferences and constraints [1, 2]. Under this view, dialogue serves mainly as a signal for improving estimates of what the user wants, which can then guide candidate retrieval, ranking, and response generation.

A direct form of this framing appears in explicit elicitation and slot-based approaches, where users are asked to specify desired attributes, constraints, or item properties directly. These systems are simple and interpretable, and they support structured user representations that downstream recommenders can consume. Clarification strategies extend this idea by asking follow-up questions when user requests are vague or underspecified [11]. In both cases, the conversational move is motivated mainly by the need to sharpen the system’s estimate of user intent.

Critiquing-based recommenders make interaction more iterative by asking users to

react to candidate items and then updating the user model accordingly [3]. Mixed-initiative and proactive dialogue systems similarly let the system decide when to ask a question, offer a suggestion, or provide additional information [12]. These methods can make recommendation more flexible and efficient, and they sometimes raise considerations the user did not mention initially. Even so, interaction is still usually framed around preference inference rather than around unresolved decision-relevant information.

This preference-centric view also appears in common user modeling schemes. CRS have represented users through symbolic preference states, latent dialogue embeddings, and natural-language summaries of preferences or constraints [13, 14, 15]. These representations differ in interpretability and flexibility, but they are typically designed to condition retrieval, ranking, or response generation on the system’s current estimate of user preferences. Conversational state is therefore more often treated as a vehicle for recommendation than as an explicit representation of decision-relevant factors.

2.4 Information-Needs in Conversational Recommendation

Preference elicitation explains only part of what conversational recommendation must accomplish [16, 17]. In realistic decision settings, users are not only expressing what they like. They are also trying to work out what they need to know before choosing among plausible alternatives. A system can therefore align well with expressed preferences while still failing to provide enough information for a well-supported decision.

Conversational recommendation is better viewed not just as preference elicitation, but as the management of information needed for choice. Under this view, *INs* extend beyond explicit factual questions. They also include partly expressed preferences, unresolved trade-offs, and other considerations that matter for choice even when the user has not stated them directly. Preference elicitation remains an important part of this process, but it does not cover the whole task.

The need for this broader view follows from a basic asymmetry in the interaction. Users bring goals and constraints, but may not fully articulate the criteria that matter. The system, by contrast, has access to catalog structure, item descriptions, and sometimes broader domain knowledge. Inferring a plausible preference profile is therefore not enough if the user still lacks the information needed to decide.

The problem is clearest when relevant attributes are latent, subjective, or hard to name in advance. Users may care about comfort or durability without being able to specify those concerns clearly at the outset [18, 5]. They may also lack the domain knowledge needed to recognize what matters for the decision or what remains unclear.

Some concerns only become salient once candidate items are presented and compared. User utterances therefore expose only part of the underlying decision problem, not the full set of INs relevant to choice.

With this perspective, we place preference elicitation within a broader account of the information a user needs in order to choose well. Existing CRS behaviors such as clarification, critiquing, and explanation do touch parts of this broader process, but they are usually framed as mechanisms for improving preference inference or recommendation quality. That leaves a different source of conversational failure under-specified: the user may still lack information needed to make their decision. Because these failures can persist even when explicit preferences have been recovered reasonably well, preference-centric evaluation alone does not capture them. An Information-Needs perspective is therefore needed to represent and assess unresolved decision-relevant information directly.

2.5 Interaction State in Multimodal CRS

Whether a CRS can support decision-making depends in part on what it represents explicitly in interaction state. In CRS, user models and dialogue state typically provide the internal representation that conditions retrieval, ranking, and response generation [1]. What is represented in state therefore determines not only what the system knows about the user, but also what aspects of the interaction it can reason about over time.

Most existing state representations are designed mainly to maintain the current estimate of the user’s preferences, constraints, or local conversational context [1]. That makes them well suited to conditioning recommendation decisions, but less naturally suited to representing and using INs. If conversational recommendation is treated more broadly as the management of decision-relevant uncertainty, preference state alone may limit the depth of engagement.

As noted in Section 2.1, MLLM systems can perform multiple conversational recommendation functions within a single interface, so distinctions that are conceptually separate can blur in practice [9, 10]. A system may therefore produce responses that sound informative and well grounded without explicitly tracking which user concerns remain open, which alternatives are under comparison, or why a particular conversational move is appropriate at that stage of the interaction. Richer interaction state is therefore useful: it can track not only preferences, but also what remains unresolved, which alternatives are salient, and what information still needs to be surfaced.

2.6 Evaluation of CRS

Evaluation shapes which CRS behaviors are optimized and rewarded. Because conversational recommenders are multi-component, interactive systems, they are difficult to evaluate holistically [16]. In practice, however, most evaluation protocols still emphasize whether the system recommends suitable items, produces fluent responses, and reaches a useful outcome efficiently.

This tendency appears in both traditional and LLM-based CRS evaluation. Common automatic measures focus on recommendation accuracy or ranking quality, while human evaluation often targets response quality, coherence, persuasiveness, or overall satisfaction [16]. Recent work has argued that such protocols can overemphasize agreement with annotated items or utterances while underemphasizing the interactive nature of conversational recommendation, including how well a system manages the dialogue over time [17].

Existing evaluation regimes do not directly ask whether an interaction helped the user obtain the information required for a supported choice. As a result, systems can perform well on surface-level metrics without actually addressing the user’s decision-relevant INs. Broader conversational-agent evaluation frameworks have begun to move toward more multi-dimensional assessment by accounting for user tasks, evaluation criteria, methodology, and stakeholder goals [19]. In multimodal CRS, however, evaluation remains underdeveloped in assessing whether an interaction actually addresses the user’s decision-relevant INs.

2.7 Toward an Information-Needs Perspective on Multimodal CRS

Taken together, the literature above shows that conversational recommendation has been developed mainly around preference modeling, recommendation quality, and response quality. Work in conversational information seeking and recent critiques of CRS evaluation, however, point to a broader concern: whether interaction helps resolve what the user still needs to know in order to choose well.

The remainder of this thesis builds on that gap. It formalizes conversational recommendation through an IN perspective and uses that perspective to study interaction quality beyond preference alignment alone.

Chapter 3

Part I: Vogue

Note: Portions of this Chapter are adapted from our prior publications *VOGUE: A Multimodal Dataset for Conversational Recommendation* [20] and *Evaluating Scene-based In-Situ Item Labeling for Immersive Conversational Recommendation* [21] with permission.

Part I of the thesis introduces VOGUE as an experimental substrate for studying multimodal conversational recommendation beyond preference alignment alone. The goal of the dataset is not only to support standard recommendation-oriented evaluation, but also to make broader aspects of conversational success more observable and thus assessable, including how dialogues surface decision-relevant questions, support comparison among alternatives, and relate conversational behavior to user satisfaction.

To support this, VOGUE was designed around a set of methodological desiderata. In addition to standard preference-alignment observables, the dataset preserves signals about decision-relevant information exchange, maintains asymmetric access to item information between Seeker and Assistant, and grounds the interaction in a genuinely multimodal setting where a distinct set of scenario-critical properties are not recoverable from visual appearance alone. Together, these design choices allow VOGUE to expose limitations of preference-centric conversational recommendation and to support the later analyses developed in this thesis.

We enumerate our desiderata below:

1. **Preference-alignment observables:** The dataset should support preference-centric evaluation, including item-level preference signals and an explicit final choice (if one was reached), so that a system can be assessed on how well it infers and aligns with surfaced Seeker choices.
2. **Decision-relevant information observables:** The conversational record should contain sufficient questions, disclosures, comparisons, and uncertainty-related be-

haviors to evaluate whether decision-critical attributes were surfaced during the interaction, rather than treating conversational success as preference alignment alone.

3. **Asymmetric information access:** The interaction should be structured such that the Seeker does not directly observe all decision-relevant item attributes, while the Assistant has access to a defined evidence pool. This creates a natural distinction between preference expression and information disclosure.
4. **Grounded multimodal uncertainty:** The task should be intrinsically multimodal, combining images and textual attributes, while ensuring that several scenario-critical properties are not reliably inferable from visual appearance alone.

We instantiate these desiderata in the fashion shopping domain. Fashion recommendation is a natural testbed because it combines (i) intrinsic multimodality, since visual appearance is central but insufficient, (ii) a mix of factual attributes (e.g., materials, care instructions) and subjective or context-dependent attributes (e.g., comfort, warmth), and (iii) substantial variation in which attributes become decision-critical across scenarios. This combination creates realistic settings where preference alignment alone is insufficient: a system may infer a user’s aesthetic preferences while still failing to surface the information needed for a well-supported choice.

To meet these desiderata, we designed VOGUE as six independent conversational recommendation trials between paired participants. We refer to the recommender as the **Assistant** and the user as the **Seeker**. Each trial centered one of 6 unique and orthogonal scenarios, giving the Seeker a temporary decision-making goal and context. The full scenarios are included in the dataset repository¹. We present all original scenarios in Appendix A.1.1 and abridged summaries next:

- **Scenario 1 (Farm visit, fall):** Outdoor farm activities in cool, potentially rainy weather with muddy ground.
- **Scenario 2 (Campus day):** A typical university day with walking between buildings in chilly, windy conditions.
- **Scenario 3 (Business casual office):** Indoor office setting with a business-casual dress code and cool temperatures.
- **Scenario 4 (Indoor social dinner):** Casual indoor gathering where appearance still matters.
- **Scenario 5 (Winter city exploration):** A full day of walking between indoor and outdoor spaces in winter.

¹The complete VOGUE dataset is available at anonymous.4open.science/r/vogue-EF02

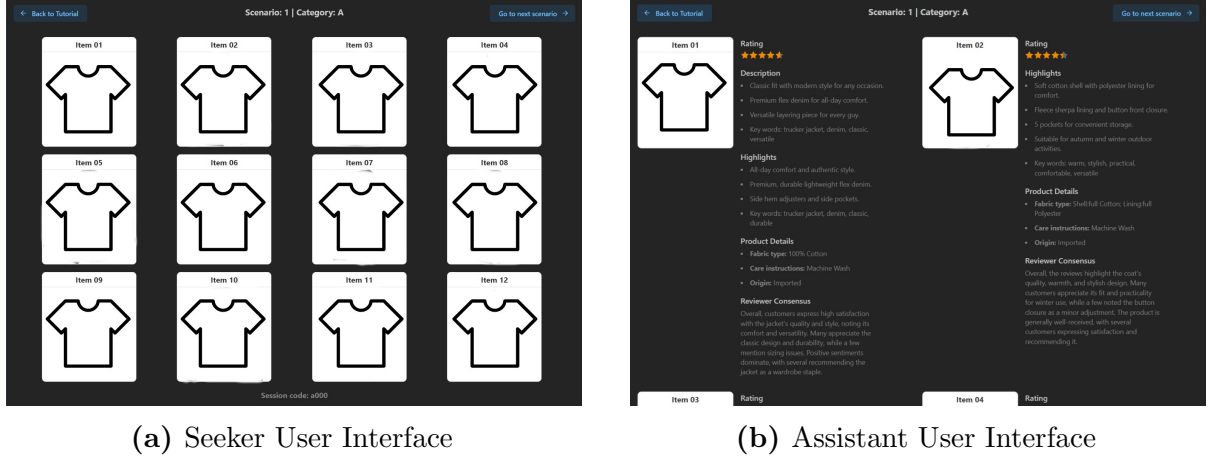


Figure 3.1: Study User Interfaces (UI) used in our experiment.

- **Scenario 6 (Park after rain, summer):** Warm weather walking through a park on potentially damp ground.

In each scenario, participants shared a catalog of 12 candidate items drawn from Amazon.ca, spanning outerwear, layering pieces, and shoes. Each item included images and collated metadata (name, description, review score, review summary). Visual grounding is intrinsic to fashion recommendation, since many critical attributes such as style, silhouette, and overall aesthetic are appearance-dependent. As a result, the task is inherently multimodal and cannot be reduced to a purely textual setting without changing its nature. At the same time, several scenario-critical properties (e.g., weather resistance, traction, care requirements) are typically not visually recoverable and must be supported by authorized textual evidence.

To reduce cognitive load, descriptions and reviews were GPT-4O-MINI summarized, while size, fit, color, and price were omitted, and participants were asked not to anchor on them.

Throughout the study, the catalog comprised 36 items (12 per scenario) plus 40 additional items used for fashion profiles. The Seeker’s task was to articulate needs and select an item, while the Assistant guided preference elicitation and made recommendations.

To make grounded information disclosure measurable, VOGUE specifies a defined evidence pool for the Assistant. During conversational trials, Assistants were restricted to the provided catalog content, including item images and collated metadata such as the provided review score and review summary, and were instructed not to use external web resources or personal ownership claims. When an attribute could not be supported from the available catalog evidence, Assistants were encouraged to treat it as uncertain rather than asserting it as fact. This protocol helps distinguish preference modeling, that is, what the Seeker values, from information disclosure, that is, what the Assistant can support from the provided evidence.

For space reasons, we present a concise description of the protocol and annotation schema in the paper, while the full experimental protocol, survey instruments, intent definitions, and annotation guidelines are provided in the repository¹.

Table 3.1: Seeker satisfaction survey items (1-5 Likert scale).

Question
• The conversation allowed me to make a good selection on what to buy. • I made a well-justified final selection. • I would actually purchase the item selected. • During the conversation, I felt in control of the process. • I found the Assistant’s recommendations helpful.

Our data collection protocol² proceeded in three steps:

1. **Onboarding surveys:** All participants completed demographic forms and fashion entrance surveys, including a profile questionnaire and pre-ratings of 40 non-candidate items, which provided baseline preferences and served as user profiles, a feature rarely included in similar datasets. (Desiderata 1, 2, 4). The complete details for all surveys are in the `surveys` folder of the supplementary material¹.
2. **Conversational trials:** In each of the six scenarios, the Seeker and Assistant discussed 12 candidate items, with the Assistant accessing full metadata and the Seeker seeing only images and the scenario description. A web-based User Interface (UI) facilitated catalogue viewing, with corresponding Seeker and Assistant views (Fig. 3.1). Assistants were instructed not to use other external resources and to elicit context through dialogue. Conversations continued until the Seeker indicated satisfaction with at least one item or rejected all options. The complete details are in the `item_ratings` subfolder of VOGUE¹.
3. **Post-conversation Surveys:** After each scenario, the Seeker provided ground-truth preference ratings of all 12 items on a 1 to 5 scale, while the Assistant attempted to predict these ratings. Participants were instructed to rate based on how likely they would be to purchase the item given the conversation and scenario, with 5 being “would purchase” and 1 being “would never purchase”. Neither party saw the other’s ratings. Both parties also completed a short feedback survey on the perceived quality of that conversation. Seekers additionally completed a 1-5 Likert scale survey of their satisfaction with the scenario and conversation. Survey questions are detailed in Tab. 3.1.

3.1 Participants

We recruited 20 participants through informal outreach at a local University and among recent graduates in the surrounding area. Young adults were recruited in order to reflect a core demographic of fashion shopping and the adoption of conversational Assistants.

Participants were randomly assigned fixed roles as Seeker or Assistant for the entire session. To prepare Assistants for their role, we conducted a short coaching session, to familiarize them with the catalog of candidate items. Seeker participants received only the task scenarios

²The protocol received approval from the University of Toronto Research Ethics Board (#00049065).

beforehand to replicate real-world conditions, necessitating that Seekers articulate their needs without access to full product knowledge.

3.2 Dialogue Intent

To support analysis of both preference-oriented and information-oriented conversational behavior, we annotate utterances with dialogue intents that capture both (i) *preference acquisition* behaviors, which elicit or refine what the Seeker values, and (ii) *information-oriented* behaviors, which request, compare, or disclose decision-relevant attributes using the available catalog evidence. We base our design on prior intent taxonomies for conversational recommendation and utterance-level labeling [22, 23], and adapt them to a fashion shopping setting with shared visual catalog and asymmetric access to item information.

Since earlier studies focused on domains outside fashion shopping and often did not include visual grounding directly in the conversational loop, existing intent sets do not cleanly separate preference refinement from information-seeking and evidence disclosure in multimodal decision settings. We therefore extend prior definitions with tags that arise naturally when users must resolve latent attribute uncertainty, including factual attribute questions, feature critiques, and explicit comparison requests. The resulting Seeker- and Assistant-side taxonomies are listed in Tab. A.1 and Tab. A.2, and the full hierarchical schema is available in the dataset repository¹.

In particular, several intent categories in our taxonomy can be interpreted as information-seeking or evidence-oriented behaviors, such as *Inquire-Factual*, *Critique-Feature*, and *Critique-Compare*, while others primarily reflect preference elicitation and refinement, such as *Provide Explicit Preferences* and *Refine Preference*. This structure supports later analyses that distinguish conversations that merely converge toward preference-aligned recommendations from conversations that also surface and address decision-critical information during the interaction.

We present a short sample exchange with tags below:

Source: c37_a_1.json | Scenario: Rainy Farm Visit

Assistant: "Since it may rain, I recommend Item02, and you could also consider Item06 and Item08, which are lighter and more stylish."
[Recommend-Show (RS), Explain-Preference (EP)]

Seeker: "In a muddy farm, waterproof matters more. Item06 and Item08 would get dirty easily. Item02 feels more rain-proof."
[Critique-Feature (CF), Refine Preference (RP)]

Table 3.2: VOGUE dataset statistics.

Statistic	Count	Statistic	Count
Conversations	60	Participants	20
Total turns	899	Total utterances	2,100
Total tokens	22,717	Total item ratings	1,440
Avg. turns / conversation	14.8	Avg. tokens / turn	36.1

Assistant: "As a Plan B, you could also consider Item07 or Item10.

Item10 is waterproof but warmer."

[Recommend-Show (RS), Explain-Additional Info (EAI)]

Seeker: "Between Item02 and Item10, I choose Item02. It's waterproof, darker, and still looks good for photos."

[Accept (ACT), Explain (EXP)]

We conducted a two-phase labeling process to annotate dialogue utterances for subsequent analysis. In the first phase, we used GEMINI-2.5-FLASH to streamline large-scale tagging. In the second phase, all tagged conversations were manually reviewed by the annotation team to correct mislabeled utterances and refine the taxonomy based on the collected dialogues. Inter-annotator reliability was assessed using Cohen's Kappa, yielding strong agreement of $\kappa = 0.81$. From these tags, we derive the intent categories used in later analyses, including categories that proxy for preference elicitation and information-seeking behaviors. For clarity, we primarily discuss overarching intent categories (e.g., *Provide* or *Explain*) rather than fine-grained tags.

3.3 Dataset Contents

VOGUE ¹ includes 60 dialogue transcripts (10 participant pairs $\times$ 6 scenarios each), transcribed and time-stamped at the turn level. Utterances are labeled with intent tags where appropriate. Transcripts are also annotated with the Seeker's final item choice and all items explicitly mentioned. Each scenario is matched to its catalogue. Items are released as utilized in the thesis with metadata. We summarize key statistics in Tab. 3.2.

3.4 Data Analysis

In this section, we examine VOGUE to better understand multimodal conversational dynamics, assess which evaluation signals are most informative for conversational success, and analyze INs beyond preference alignment alone.

Specifically, we organize the analysis around three complementary research questions. First, we examine how shared visual grounding shapes the dynamics and stages of dialogue in an

inherently multimodal domain. Second, we study several candidate success signals in VOGUE, including human preference alignment, Seeker satisfaction, and offline MLLM preference recovery. Third, we analyze inherited ICRS-style IN evaluation in order to motivate a broader IN perspective for multimodal CRS.

To this end, we consider the following three research questions for Part I:

- RQ1: Multimodality dynamics.** How does visual grounding alter the dynamics and stages of dialogue, compared to text-only settings? In an inherently multimodal domain like fashion, what structures and processes emerge, and how should CRS adapt to these?
- RQ2: Success signals.** To what extent do post-conversation item-level preference ratings predicted by human Assistants and modern MLLMs align with the Seekers’ ground-truth preferences? Further, to what extent does such preference alignment correspond to the subjective satisfaction of Seekers?
- RQ3: Information-Needs.** To what extent do IN criteria capture aspects of conversational success beyond explicit preference alignment? More generally, how do these results inform our understanding of INs in multimodal CRS?

3.4.1 RQ1: Multimodality Dynamics

We examine how visual grounding shapes the structure of multimodal recommendation dialogue. In particular, we analyze when items are introduced, how initiative shifts between participants, and how recommendation, critique, and refinement unfold over the course of an interaction. To do so, we study the distribution of conversational intents, item mentions, and intent-tag transitions across each dialogue, with selective comparison to related text-only settings where relevant. For clarity, we omit the *Acknowledgment* and *Others* tags, as they do not contribute meaningfully to task progression.

As shown in Fig. 3.2, Assistants consistently introduce new items in two distinct waves: the first following initial preference and context elicitation, and the second after refinement based on earlier feedback. We also observe that Assistants frequently group recommendations, presenting up to four items simultaneously for discussion. This differs from the more linear, sequential item presentation often seen in text-only conversational settings. Overall, the conversations exhibit a recurring staged structure in which visual grounding supports comparison, critique, and refinement over sets of concrete candidates rather than isolated items. The distributions in Fig. 3.2 further suggest a clear stage-like structure: activity is clustered into distinct peaks rather than being evenly spread across the dialogue, with a characteristic ebb and flow as initiative flips between participants. We therefore segment the conversations into five stages, each defined by the dominant (i.e., most prevalent) intents and role dynamics at that point. In Fig. 3.3, we provide an illustrative example of each stage based on conversations in VOGUE:

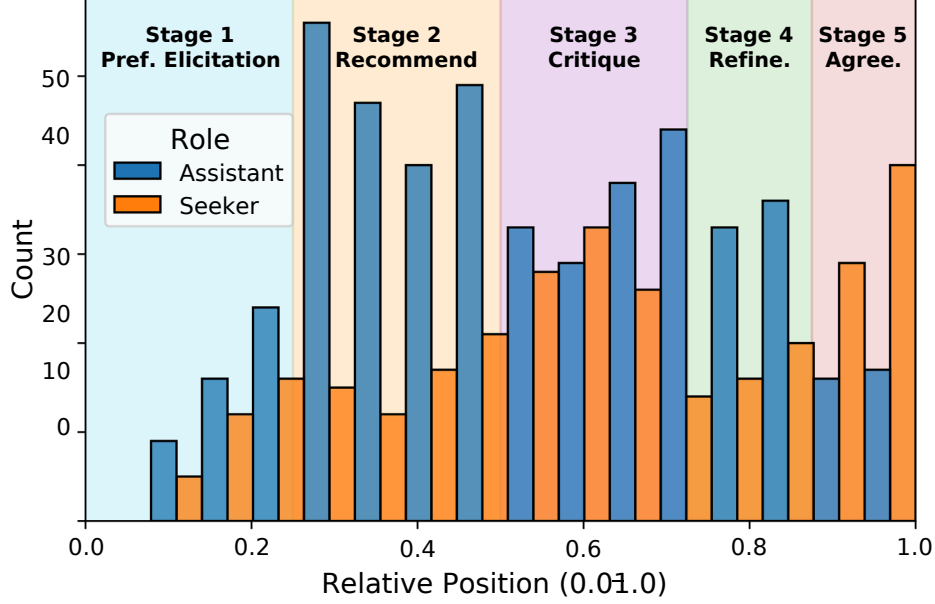


Figure 3.2: Seeker & Assistant Item Mention Distribution by Conversation Proportion. We observe a bimodal distribution in the Assistant, corresponding to *First Recommendation* and *Refinement* stages respectively.

Stage 1: Preference Elicitation. Early exchanges are dominated by preference elicitation (Assistant *Request*; Seeker *Provide*, *Ask for Recommendation*, and *Answer*).

Stage 2: Recommendation. This is followed by the first recommendation wave (Assistant *Recommend*), with *Explain* rising soon after. Interestingly, we observed that participants often discussed the items in groups according to abstract unifying properties of those item groups.

Stage 3: Critique. The conversation then transitions to a bridge characterized by Seeker *Critique*, *Inquire*, and evaluative ratings, while Assistant *Explain* and *Personal Opinion* remain high.

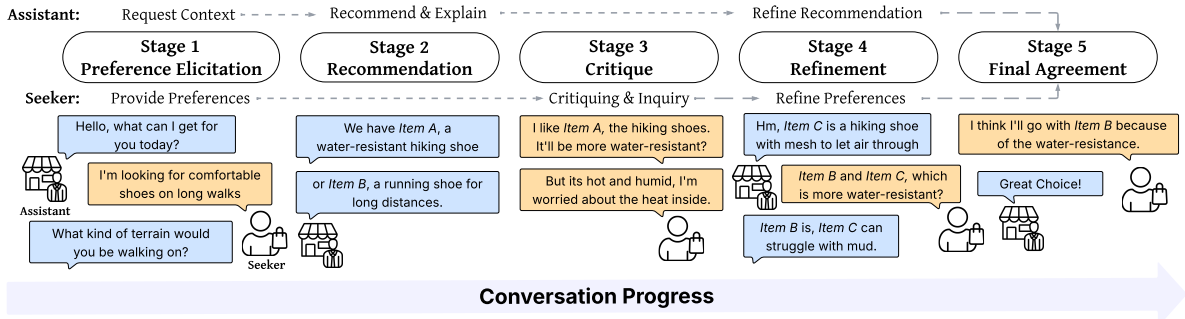


Figure 3.3: Our proposed stage splits, illustrating Recommender–User and Seeker–Assistant initiative and major intent tags. Examples of dialogue are included, along with each role’s motivation at each stage.

Stage 4: Refinement. Stage 4 is strongly characterized by Assistants introducing a new, reduced set of items (usually one or two) to refine the recommendation based on Seeker critiques. Concurrently, Seekers refine their preferences based on available attributes across recommended items, continuing to utilize property grouping as observed in *Stage 2*.

Stage 5: Final Agreement. Finally, Stage 5 is marked by Seeker critique and *Recommendation Rating* peaking, particularly *Accept* tags, signaling choice and resolution.

Significance to CRS. Our analysis indicates that visual grounding and multimodality reshape conversational dynamics relative to text-only CRS settings. Dialogues in VOGUE tend to follow a recurring staged structure, and shared images allow participants to discuss concrete candidates compactly without requiring exhaustive verbal description. Conversations average 14.8 turns and 36 tokens per turn, yet still progress through elicitation, recommendation, critique, refinement, and final agreement.

Inspection of the transcripts further suggests frequent use of positional references and visually grounded stylistic shorthand when discussing catalogue items. In contrast to settings without a shared visual catalogue [22], this gives participants a common frame of reference for discussing multiple items and their distinguishing attributes. Recommendation therefore often proceeds not only through direct preference elicitation, but also through comparison, critique, and narrowing over small groups of candidates.

We also observe that Assistants often avoid simply continuing to produce new recommendations. After the first batch of items is introduced, many conversations enter a distinct critique stage in which those initial recommendations are used to narrow, revise, and clarify the decision. In this phase, the recommended set becomes a shared object of reasoning for both parties. The critique stage centers less on item-level preference feedback than on clarifying what remains unresolved in the interaction. Assistants may use critique to elicit further explanation of the Seeker’s decision criteria, while Seekers may infer gaps or misunderstandings in the Assistant’s earlier framing from the recommendation choices themselves. The exchange therefore often involves implicit reasoning over unresolved informational gaps, including what still needs to be established for the decision and what each party takes the other to know, misunderstand, or still need. The decision-relevant information surfaced in this stage motivates the INs formalized later.

Taken together, these observations suggest that multimodal CRS should be understood as interaction over recommendation sets, rather than as text-based recommendation with images attached. In domains such as fashion, those sets often provide the basis for comparison, critique, and refinement across turns, while also helping bring unresolved criteria into the conversation. Conversational progress therefore depends both on preference elicitation and on decision-relevant information exchange.

3.4.2 RQ2: Success Signals in Vogue

We next examine several complementary success signals available in VOGUE. Preference alignment captures how well an Assistant recovers the Seeker’s underlying item-level preferences from the dialogue, while post-task satisfaction captures how the interaction was experienced by the Seeker. We then extend this view with an inherited IN evaluation from prior ICRS work, which operationalizes whether the interaction surfaces information relevant to explicit intent and broader decision-relevant needs. Finally, we consider offline MLLM-based preference inference to assess how well strong modern models can recover Seeker preferences from the same conversational evidence.

Taken together, these analyses provide several distinct lenses on conversational success. Some remain preference-centric, such as rating alignment. Others are more interaction-centric, such as satisfaction and IN support. Having motivated this broader perspective earlier, we move to examine them here.

Preference Alignment Between Human Assistants and Seekers. We first evaluate the quality of human Assistant interaction in VOGUE using quantitative alignment between human Assistant predictions and Seeker ground-truth ratings. Following Jannach et al. [1] and McNee and Riedl [24], we treat alignment as recovery of both the structure and intensity of preferences rather than simple accuracy, and report Pearson Correlation (PC) and Mean Absolute Error (MAE) between item-level ratings.

We observed that Seekers strongly favored low scores (44% 1s, only 8.6% 5s), resulting in a highly imbalanced distribution. Human Assistants nonetheless achieved consistently good preference alignment, with a mean PC of 0.64 (median 0.61, SD 0.21) as shown in Fig. 3.4, indicating reliable recovery of preference directions rather than isolated success cases. MAE, however, varied widely across scenarios, from as low as 0.25 to nearly 2.0, underscoring persistent challenges in modeling preference strength.

Since each Seeker engaged with the same Assistant in six independent scenarios, we also analyze performance across repeated interactions. PC improved and MAE declined over successive scenarios, suggesting that Assistants became more effective with practice, despite neither party ever seeing the other’s item ratings. While improvement trends were modest, the results highlight the importance of long-term user modeling in conversational recommendation settings and indicates potential for adaptive learning without explicit item-level feedback.

Relationship Between Alignment and Seeker Satisfaction. We also record Seeker satisfaction per conversation. In Fig. 3.5, we observe that Seeker satisfaction was consistently high. Likert surveys captured perceived decision quality, justification, intent to purchase, and Assistant helpfulness. Most scores exceeded 4.5 (out of 5), with helpfulness and justification especially positive.

We further analyze the relationship between Assistant alignment and Seeker satisfaction

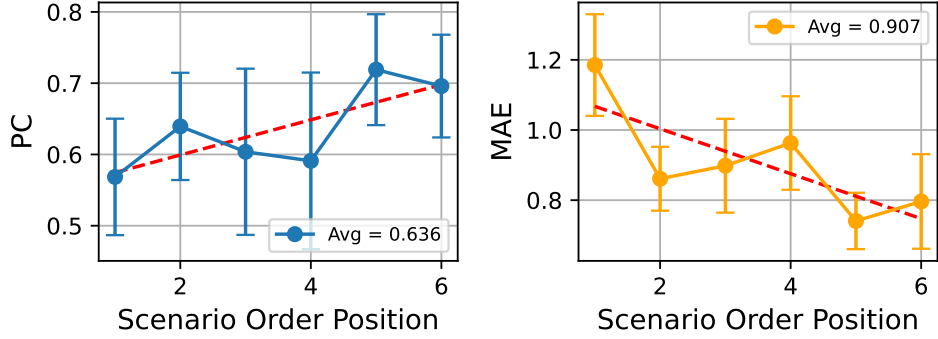


Figure 3.4: MAE↓ and alignment metrics as a function of scenario order position. Scenario order is the order in which individual trials were conducted for each Assistant–Seeker pair.

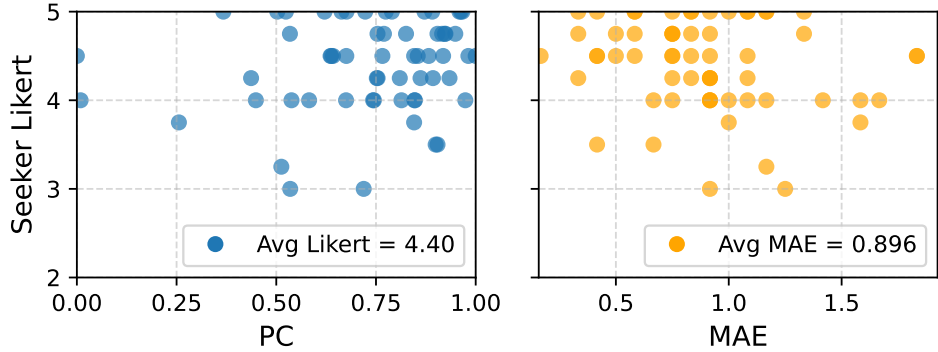


Figure 3.5: Relationship between Seeker satisfaction and alignment metrics. PC (left) shows strong clustering with higher satisfaction, while MAE (right) reveals broader variability.

(cf. Fig. 3.5). Encouragingly, higher satisfaction generally coincides with stronger alignment, measured by both PC and MAE, indicating that better alignment contributes to better user experiences. This strong correlation suggests that alignment metrics may serve as a rough proxy when direct user satisfaction measurements are unavailable.

Nonetheless, satisfaction varied across perceived decision impact and purchase intent, underscoring that it is not monolithic. This suggests that measuring only alignment, as is common in existing datasets, is insufficient. By contrast, VOGUE demonstrates how robust preference modeling and user-centered design can work together in CRS. Consistent with prior research [24], systems that feel supportive and transparent, while also aligning with user preferences, are more likely to elicit user satisfaction and trust.

Offline MLLM Preference Inference from Full-Conversation Evidence. Having established the impact that alignment can have on user experience, and the difficulties LLMs encounter with proactive understanding in CRS, we next examine how modern MLLMs perform on the same preference recovery task as in VOGUE. Here, the goal is not to evaluate end-to-end dialogue behavior, but to assess a model’s ability to infer Seeker ratings from a completed

Table 3.3: Alignment metrics and accuracy for human Assistants and MLLMs. Best results from models and Assistants are **bolded**, second-best are underlined. Arrows indicate whether higher or lower is better. *PC undefined (uniform rating, zero variance)

Model	Acc.↑	PC↑	MAE↓	M-MAE↓	MAE by GT↓				
					GT=1	GT=2	GT=3	GT=4	GT=5
Human Assistants	0.397	0.636	0.896	0.849	0.746	<u>0.891</u>	<u>1.171</u>	<u>1.284</u>	0.597
GPT-5-MINI	0.344	<u>0.083</u>	1.296	1.543	1.041	0.845	1.209	2.222	2.516
GPT-4O-MINI	0.156	0.000	1.619	<u>1.491</u>	2.160	1.186	0.953	1.160	<u>1.726</u>
GEMINI-2.5-FLASH	0.406	0.057	<u>1.356</u>	1.681	<u>0.853</u>	0.930	1.628	2.444	2.839
GEMINI-2.0-FLASH	<u>0.402</u>	0.041	<u>1.356</u>	1.772	0.693	1.031	1.736	2.568	3.065
Baseline: Random	0.240	-0.005	1.713	1.730	1.997	1.403	1.147	1.469	2.387
Baseline: Mode	0.443	NaN*	1.219	1.932	0.000	1.000	2.000	3.000	4.000
Rating									

conversation given the full item catalogue, metadata, and transcript. Vitally, only a few items are discussed, understanding of implicit signals is paramount.

All models are provided with the same prompt (provided below), the full item catalogue and metadata, and the complete tagged transcript at the conversation level. They are then asked to complete the same item-rating survey task as the human Assistants. We compare these models against both human Assistants and two simple baselines: a random baseline that samples from the empirical Seeker rating distribution, and a mode baseline that predicts the most common rating of 1.

You are an expert sales associate and fashion expert, with deep experience in apparel, styling, sales, and customer relations, reviewing a recorded conversation between an assistant and a seeker. The assistant provides shopping advice and help to the seeker, who is attempting to make a clothes purchase. You must rate each of the 12 items on a 1-5 scale (inclusive).

Criteria: Rate the items based on how likely you think the Seeker is to purchase them, given the conversation and scenario that just concluded.

<format guidelines>

<catalogue, metadata and conversation history here>

We report PC and MAE for all evaluated models in Tab. 3.3, and also include Macro MAE and per-class MAE to account for the strong class imbalance. Across nearly all metrics,

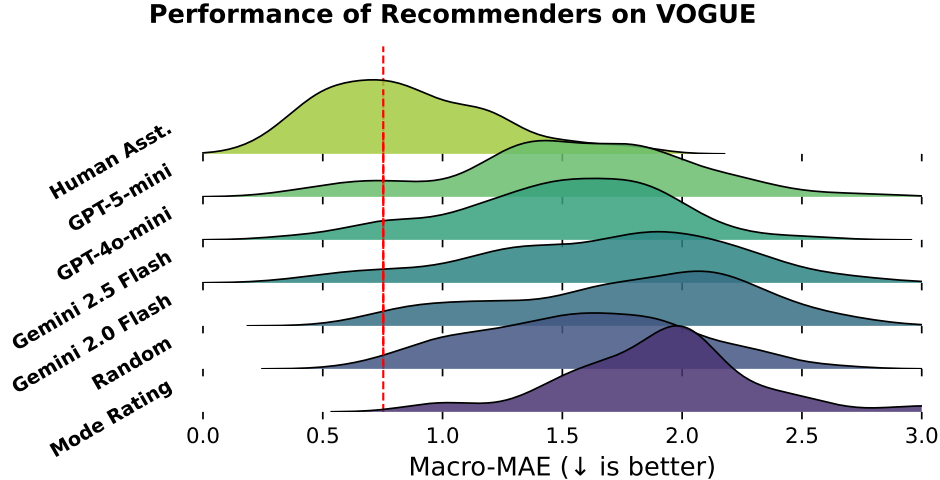


Figure 3.6: Per-class MAE across evaluated models, highlighting class-specific calibration failures under strong label imbalance.

human Assistants remain best or second-best. No tested MLLM is competitive with real human recommenders on the full preference recovery task (Fig. 3.6).

Examining predicted rating distributions reveals systematic failures of calibration (Fig. 3.7). Different model families distort the ground-truth distribution in different ways, but all perform poorly in PC relative to human Assistants. This directly undermines their ability to recover natural human preference structure. The problem becomes even clearer when considering preference generalization over the full catalogue. Since many items were never explicitly discussed, effective recommendation requires models to generalize from the dialogue to unseen candidates. Here too, MLLMs lag substantially behind human Assistants, and some perform worse than uninformed baselines.

Viewed only through raw accuracy, several MLLMs appear competitive. However, this is largely an artifact of class imbalance and the relative ease of recovering extreme negative ratings. A trivial strategy that predicts all 1s outperforms all evaluated MLLMs in accuracy. Accordingly, accuracy is not a meaningful primary metric for preference alignment in VOGUE, and we instead treat PC as the headline measure.

Thus, extending preference inference beyond explicit conversational cues remains an open challenge for current MLLMs. Even with access to the full catalogue and completed transcript, we do not observe strong evidence that these models form a coherent preference model of the Seeker.

Since VOGUE also places the interaction in a setting where decision-critical attributes are not directly available to the Seeker and instead must be communicated during the dialogue, strong preference alignment remains only a partial account of success. Interactions must also resolve relevant INs. Thus, preference-centric success signals, though useful, are insufficient alone, and a broader Information-Needs perspective is warranted.

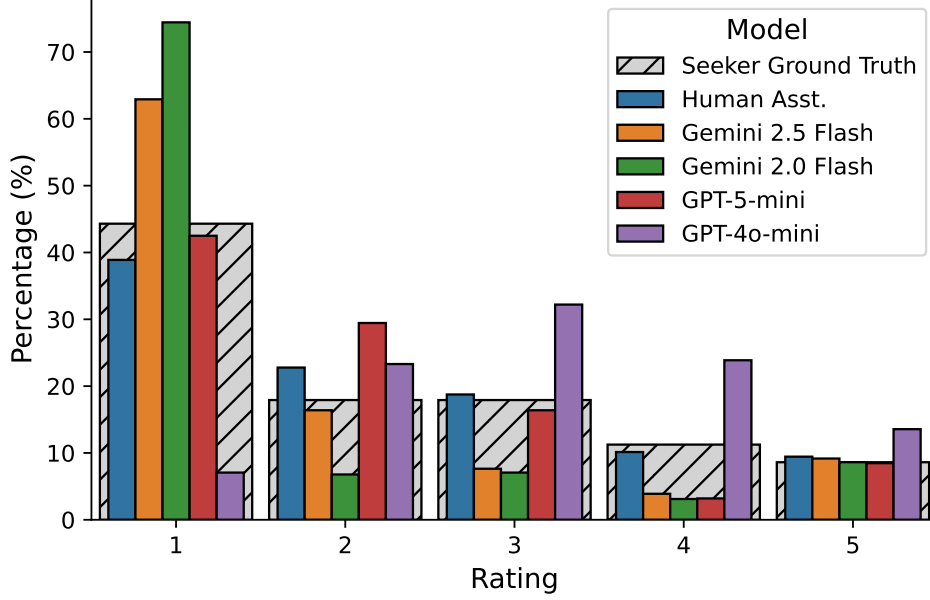


Figure 3.7: Predicted rating distributions across human Assistants, baselines, and evaluated MLLMs, compared against the ground-truth Seeker distribution.

3.4.3 RQ3: ICRS Information-Needs

As the earlier discussion of RQ1 suggests, conversational progress can depend on decision-relevant information. Assistant behavior already reflects this, especially during critique and refinement after the first recommendation set is introduced. To study these INs more directly, Liang et al. [21] introduced an IN-oriented evaluation in an immersive CRS setting. That evaluation is narrower than the interaction-level account developed later in this thesis, but it provides a tractable offline measure of whether the interaction surfaces decision-relevant information beyond explicit preference alignment alone.

The inherited setup evaluates *immersive label selection*, where a system must select which non-visual item attributes should be surfaced for recommended items given a conversation prefix $\mathcal{C}_{1:t}$. It distinguishes two levels of user intent. *Explicit Intent Satisfaction (EIS)* evaluates the ability to provide surface-level justification for the recommended item by measuring how well the selected atomic attributes address the seeker’s explicitly stated requests and preferences in $\mathcal{C}_{1:t}$. *Information Need* evaluates a system’s ability to infer the seeker’s INs from two complementary CRS perspectives: *IN-S*, alignment with information the seeker is likely to request in subsequent turns, and *IN-E*, alignment with information a knowledgeable recommender would infer as necessary to meet the seeker’s needs. [21]

Ground truth is constructed offline by dividing the full conversation $\mathcal{C}_{1:T}$ into a system-observable prefix $\mathcal{C}_{1:t}$ and an unobserved suffix $\mathcal{C}_{t:T}$ (Fig. 3.8). Each utterance is annotated with zero or more criterion-specific intent tags:

1. *Explicit Seeker Request*: seeker utterances in $\mathcal{C}_{1:t}$ that express explicit preferences, for which

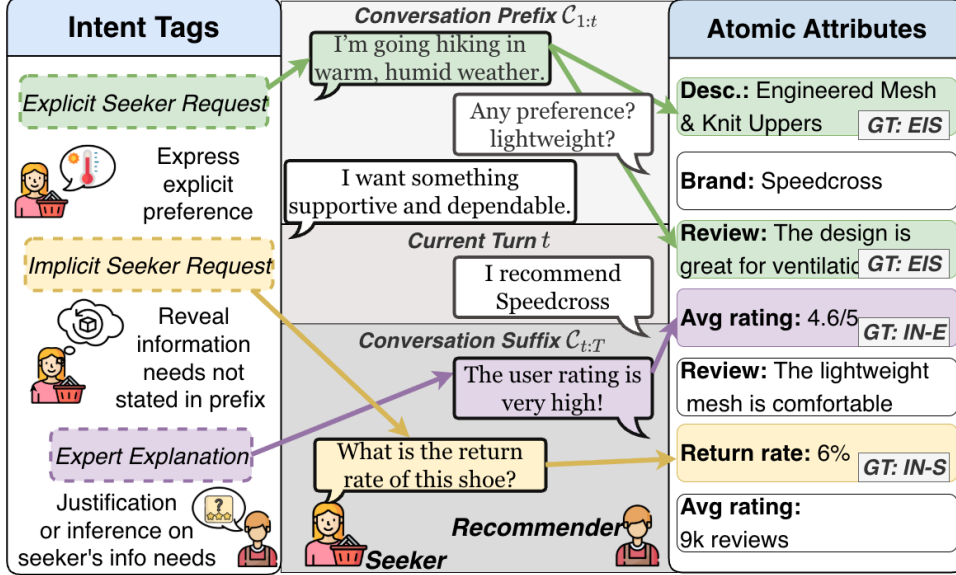


Figure 3.8: Ground Truth for Label Selection. Ground truth for the three proposed criteria is constructed by assigning intent tags (left) to each utterance in the full conversation (center), and selecting atomic attributes that satisfy at least one utterance (right) as ground truth.

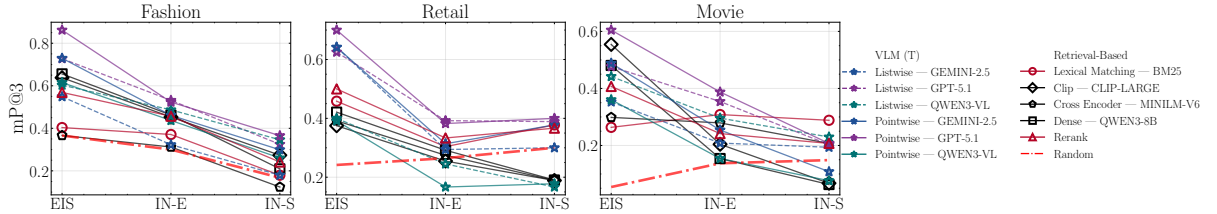


Figure 3.9: Performance of existing immersive label selection methods under three defined criteria across scenarios. While methods achieve strong performance on *EIS*, performance degrades substantially on *IN-E* and *IN-S*.

addressing them satisfies *EIS*.

2. *Implicit Seeker Request*: seeker utterances in $C_{t:T}$ that reveal INs not stated before turn t , for which addressing them satisfies *IN-S*.
3. *Expert Explanation*: recommender utterances in $C_{t:T}$ that provide justifications for recommendation or inferences about the seeker's needs, for which being relevant to them satisfies *IN-E*.

A candidate attribute is considered relevant to a criterion if it semantically addresses at least one utterance in the corresponding criterion-specific set and is not directly inferable from the item's visual appearance.

The original ICRS paper [21] evaluated INs across 3 independent recommendation datasets, namely VOGUE, CoSRec [25], and an adapted merge of ReDial [14] and INSPIRED [26], however we will focus primarily on VOGUE-specific or common results. In Fig. 3.9, both VLM-based and several retrieval-based methods achieve high mP@3 on *EIS*, which suggests that they can reliably

identify and address explicit seeker preferences and constraints. In contrast, performance drops substantially across methods for both *IN-E* and *IN-S*. The degradation is most pronounced for *IN-S*, where several methods perform comparably to a random top-3 baseline. These results indicate that existing methods struggle not only to align with expert judgments, but also to capture the more challenging latent proactive level of INs.

Prior ICRS analysis further shows that performance improves substantially when systems are given later utterances tagged as *Implicit Seeker Request* and *Expert Explanation*, reducing the task from inferring unstated needs to matching explicit requests more directly. This gain indicates that a major source of difficulty lies in accounting for INs that are not yet explicit in the current dialogue prefix. Taken together, these inherited results suggest that explicit intent is much easier to satisfy than broader and more proactive INs.

Limits of Preference-Centric Success Signals. Taken together, the Part I analyses show that preference-centric accounts of conversational success are incomplete for multimodal recommendation. In VOGUE, conversations regularly develop through comparison, critique, and refinement over recommendation sets, which often bring unresolved criteria into the interaction. Additionally, human Assistants in VOGUE explicitly utilized INs to shape and progress recommendation. The inherited ICRS results reinforce the same pattern by showing that explicit intent is substantially easier to satisfy than anticipatory information support. Conversational success therefore depends not only on recovering surfaced preferences and constraints, but also on how unresolved decision-relevant information is brought into and managed through the interaction. This motivates the IN formalization developed next.

3.4.4 Formalizing Information-Needs

Motivation from Vogue. The earlier analyses, especially the critique-stage patterns in VOGUE, point to a broader interaction process than preference matching alone. Current CRS analysis still has limited vocabulary for describing how decision-relevant information is surfaced and managed over the course of the interaction. We therefore formalize that process here in terms of Information-Needs.

Information-Needs as question–answer requirements. To capture this, we define an Information-Need (IN) as a *decision-relevant question–answer requirement* whose resolution would materially reduce uncertainty bearing on the user’s choice. The requirement need not appear as a literal question in the dialogue. It may instead be inferred from what the user has already revealed, from the comparison currently underway, or from relevant evidence that has not yet been surfaced.

This is an analytic representation rather than a claim that users internally formulate all needs as explicit questions. The point is to place heterogeneous decision-relevant requirements into a common form based on what would need to be resolved for a supported choice.

For example, if a user says they prefer wool over cotton, the relevant requirement may be: “Is the candidate item made of wool?” If prior context suggests that weather protection matters, the relevant requirement may be: “Is the candidate item waterproof?” If the user is comparing two candidates and appears uncertain, the relevant requirement may be comparative rather than item-specific, such as: “Which option is warmer?”

From item-level to interaction-level IN. Our earlier ICRS work introduced a distinction between explicit and broader INs in an item-level label-selection setting. There, the task was to determine which non-visual attributes should be surfaced for a recommended item given a fixed dialogue prefix $C_{1:t}$. In that setting, the temporal reference point is comparatively simple: the system observes a prefix, and the evaluation asks which attributes should be selected relative to that prefix.

In multimodal conversational recommendation, however, the relevant unit is broader than a single item and broader than a single static recommendation point. A need may concern a comparison between items, an unresolved decision criterion, a likely follow-up question, or a domain-specific consideration not yet raised by the user. More importantly, such needs are tied to an interaction that evolves turn by turn. The relevant reference point is therefore not a fixed prefix, but the current state of an ongoing conversation.

Temporal status in interaction. This creates a key temporal difference from the earlier item-level formulation. At the interaction level, a need may initially be *proactive* relative to the current turn, but later become *retroactive* once the conversation makes it interactionally active. Status is therefore not a permanent property of the need in isolation. Rather, it is defined relative to a moving conversational reference point.

Because the dialogue evolves, proactive and retroactive status must be interpreted dynamically. A requirement may be proactive at turn t , i.e. not yet interactionally active but likely to surface, and then become retroactive at turn $t + 1$ or $t + k$ once the user asks about it, implies it through critique, or the recommender introduces evidence that makes it salient. Retroactive needs are thus those that have already become active by the current turn, even if they were previously only candidates for anticipation.

Two-axis formalization. We organize INs along two dimensions: *source* and *status*. Source distinguishes *user-sourced* needs from *expert-sourced* needs. User-sourced needs are grounded primarily in what the user has revealed or implied about their goals and constraints. Expert-sourced needs arise from considerations supported by catalog evidence or domain knowledge that the user may not know to request explicitly.

Status distinguishes whether a need is already *retroactive* at the current turn or remains *proactive*. Retroactive needs are already interactionally active: they have been made relevant by something established in the dialogue, whether through an explicit question, a stated preference, a comparison under discussion, contextual information already revealed, or a system disclosure

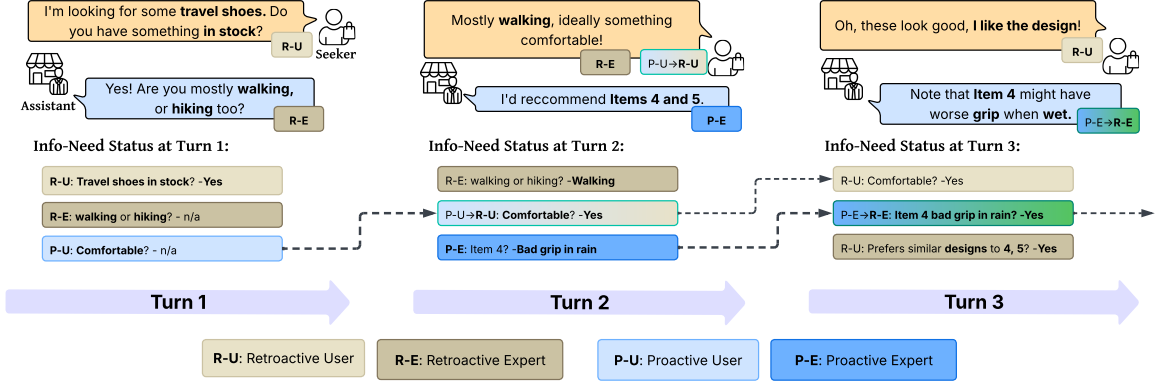


Figure 3.10: Dynamic status of INs over conversation turns. IN status is defined relative to a moving conversational reference point: a need may be proactive at one turn, then later become retroactive.

that has made the issue salient. Proactive needs are not yet active at the current turn, but are plausible candidates for anticipation or disclosure.

This decomposition refines the earlier ICRS distinction. Explicit Intent Satisfaction (EIS) can be viewed as a narrower case of retroactive, user-sourced need, namely information made active through explicit user requests, preferences, or constraints. The broader IN category is unpacked here into the four cases shown in Table 3.4.

At turn t , let $\mathcal{Q}_t$ denote the set of decision-relevant question–answer requirements under consideration relative to the current conversational state. We represent each requirement as $q = (z, \alpha, \sigma)$, where $z \in \{U, E\}$ denotes source, $\alpha \in \{R, P\}$ denotes status at turn t , and σ denotes semantic content. This yields four subsets:

$$\mathcal{Q}_t^{RU}, \quad \mathcal{Q}_t^{RE}, \quad \mathcal{Q}_t^{PU}, \quad \mathcal{Q}_t^{PE}.$$

These categories are not claims about fixed mental states, but an analytic decomposition of what kinds of decision-relevant requirements are active or worth anticipating at a given point in the evolving interaction. We present a demonstrative progression of INs in a conversation via Fig. 3.10, and a set of examples in Tab. 3.5.

Status	Source	
	User	Expert
Retroactive	<i>R-U</i> ($\mathcal{Q}^{RU}$): A question already made interactionally active by the user’s expressed preferences, constraints, questions, comparisons, or other dialogue signals.	<i>R-E</i> ($\mathcal{Q}^{RE}$): A question already made interactionally active by the dialogue context, but whose importance is inferred from domain knowledge or catalog evidence rather than explicit user formulation.
Proactive	<i>P-U</i> ($\mathcal{Q}^{PU}$): A question not yet active in the interaction, but likely to become relevant given the user’s inferred goals, values, prior behaviour, or earlier dialogue signals.	<i>P-E</i> ($\mathcal{Q}^{PE}$): A question not yet active in the interaction, but worth surfacing because domain knowledge or catalog evidence suggests it may materially affect the choice.

Table 3.4: Two-dimensional taxonomy of INs in multimodal conversational recommendation.

Label	Dialogue situation	Q formulation	Rationale
R-U	The user says they only buy cruelty-free products.	Does either candidate violate the user’s cruelty-free constraint?	The constraint is explicitly stated by the user and is already active in the interaction.
R-U	The user asks which of jackets A and B is warmer.	Which jacket, A or B, is warmer?	The comparison is explicitly requested by the user.
R-E	The user says they will be walking all day at an outdoor event, but never mentions comfort directly.	Which candidate is more comfortable for prolonged walking?	The dialogue context makes comfort relevant, even though the user did not formulate it explicitly.
R-E	The user says they want something suitable for frequent travel.	Which candidate is easiest to pack, maintain, and re-wear across settings?	The need is made relevant by the surfaced context, but its importance is inferred from domain knowledge about travel use.
P-U	The user has repeatedly rejected items with certain materials, but has not asked about the current candidate.	Does the current candidate satisfy the user’s likely material preference?	The user has not asked yet, but earlier behaviour suggests this will likely be relevant.
P-U	Earlier turns suggest the user strongly prefers close-fitting items, but fit has not yet been discussed for the current candidate.	Is this candidate slim fit?	The need is not yet active, but likely to arise given prior user signals.
P-E	Product evidence indicates that the current candidate tends to bleed dye after washing.	Does this item have a known dye-transfer issue?	The user may not know to ask, but available evidence suggests the issue may affect the decision.
P-E	The current candidate has unusually demanding care instructions.	Is this item difficult to maintain?	The user has not raised the issue, but product evidence suggests it may materially affect suitability.

Table 3.5: Illustrative reformulation of INs as question–answer requirements. Categories are determined by the source of the need and whether it is already active in the interaction or remains proactive, not by whether it was originally expressed as a literal question.

Chapter 4

Part II: MLLMs for Multimodal CRS

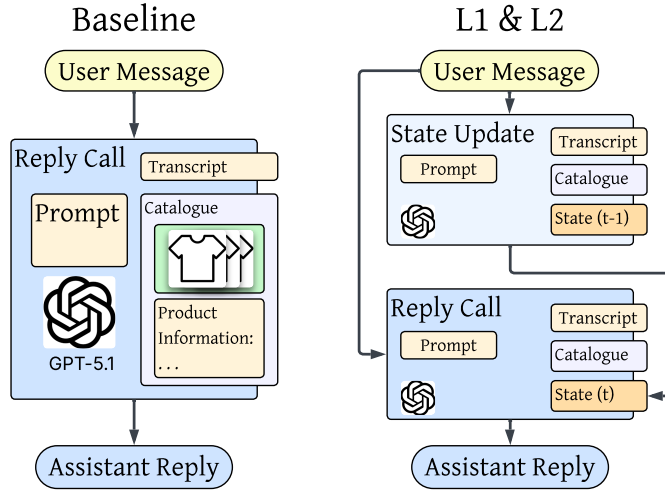


Figure 4.1: Design progression from the unstructured baseline to L1 and L2. All systems share the same live test bed and catalogue setting; the main difference is the amount of persistent interaction state explicitly maintained across turns.

4.1 Methodology

Part II introduces a test bench for studying multimodal large language model (MLLM)-driven conversational recommendation in a controlled catalogue-shopping setting. Rather than conducting a second formal user study, we built a demonstration system that reuses the core VOGUE task structure while replacing the human recommender with an MLLM chatbot. The goal is to expose recurring interaction failures and the design pressures they create. This complements Part I: whereas VOGUE analyzed success in multimodal recommendation, Part II examines where strong current MLLM systems break down in practice. Accordingly, Part II is best understood as a structured live evaluation environment, with an emphasis on understanding interaction

quality.

In total, we ran approximately 25 conversations, primarily with the author and peers, using the same fashion-shopping task format as VOGUE. Sessions were manually reviewed in full, with analysis centered on interaction patterns that are difficult to capture through aggregate metrics alone. Quantitative summaries are presented in support of these observations.

Task protocol. To preserve continuity with Part I, the live demo follows the same overall task structure as VOGUE. Users act as Seekers in a fashion recommendation scenario drawn from the same scenario pool, and each session is grounded in the same three 12-item catalogues.

Relative to the earlier user study, the main change is that the human Assistant is replaced by an MLLM-based recommender (GPT-5.1). The task otherwise remains aligned with VOGUE, preserving a fixed candidate set, shared visual grounding, and multi-turn dialogue.

Post-session data collection also mirrors the VOGUE protocol. After each conversation, users complete the same follow-up assessments, including item-rating surveys and satisfaction-oriented survey measures.

Interface and deployment test bench. The live demo was implemented as a local interactive test bench with a lightweight web interface and a FastAPI backend. Figure 4.2 shows the interface used throughout Part II. The interface was designed to closely mimic the structure of VOGUE while making live interaction practical. The left side presents the full 12-item catalogue as a persistent visual grid, with each item displayed using an image and item identifier. The right side provides a standard chat window in which the user interacts with the Assistant over multiple turns. Top-level controls are mainly logistical and not essential to conversation progression.

4.2 Systems and design progression

This section introduces the three system conditions compared in Part II: the unstructured baseline, L1, and L2. All three are deployed in the same live test bench described above and share the same interface, task format, model family, and catalogues. The differentiating factor is what interaction state is explicitly maintained across turns.

Figure 4.1 summarizes the design progression. The baseline provides a strong but minimally structured MLLM reference point. L1 adds lightweight persistent state intended to improve continuity, preference persistence, and coarse stage control. L2 extends L1 with VOGUE-motivated additions, namely explicit IN tracking, richer dialogue observations, and recommendation-set history.

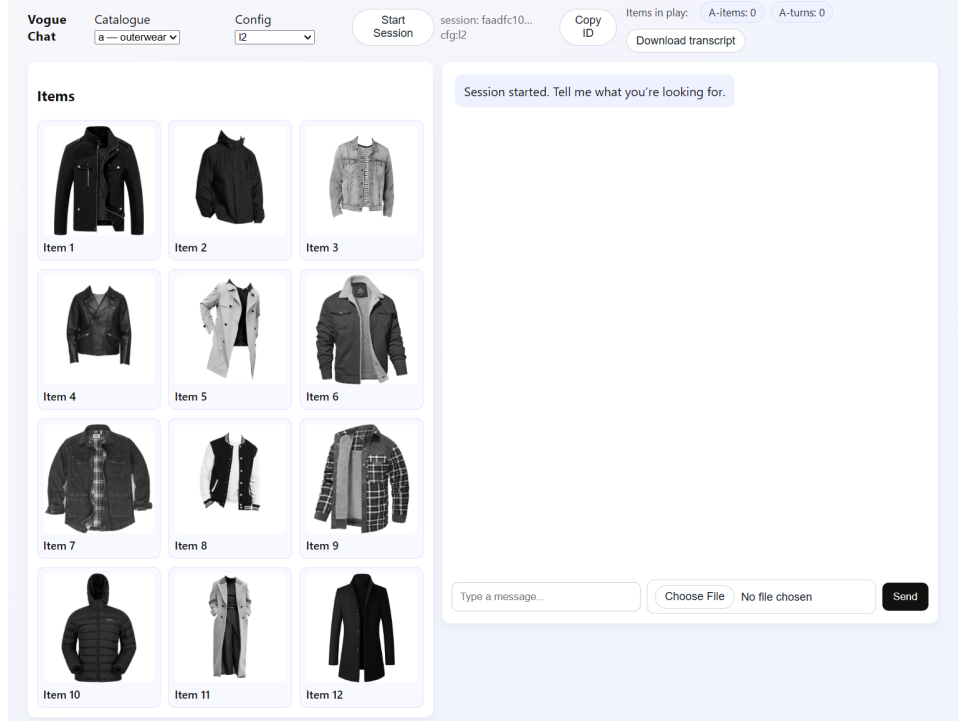


Figure 4.2: UI used for the Part II live demo. The left panel displays the shared 12-item visual catalogue, while the right panel supports multi-turn chat with the MLLM recommender.

4.2.1 Baseline

The baseline is the simplest operational instantiation of the test bench. We use the same MLLM, GPT-5.1, for all model calls, with default temperature and reasoning settings. The model is given the current conversation history together with the full 12-item catalogue context, including item metadata and the shared catalogue image. Because the candidate set is small and fully available in context, and to reduce the effect of retrieval or tool-use variation, we did not rely on external retrieval, top-k rankers, or similar tools in the deployed baseline.

Conceptually, the baseline is intended to reflect a strong but minimally structured MLLM recommender. It has access to the same dialogue and visual evidence as the later variants, but it does not explicitly maintain interaction state beyond the default per-turn prompting pipeline. In particular, it does not preserve richer preference and constraint information, recommendation-set history, or unresolved decision-relevant information across turns. This makes it an appropriate reference point for isolating what later structured additions contribute.

4.2.2 L1: Lightweight preference and policy state

L1 serves as a literature-grounded intermediate condition between the unstructured baseline and the richer interaction-state design introduced later in L2. It is adapted from *Retrieval-Augmented Conversational Recommendation with Prompt-based Semi-Structured Natural Language State*

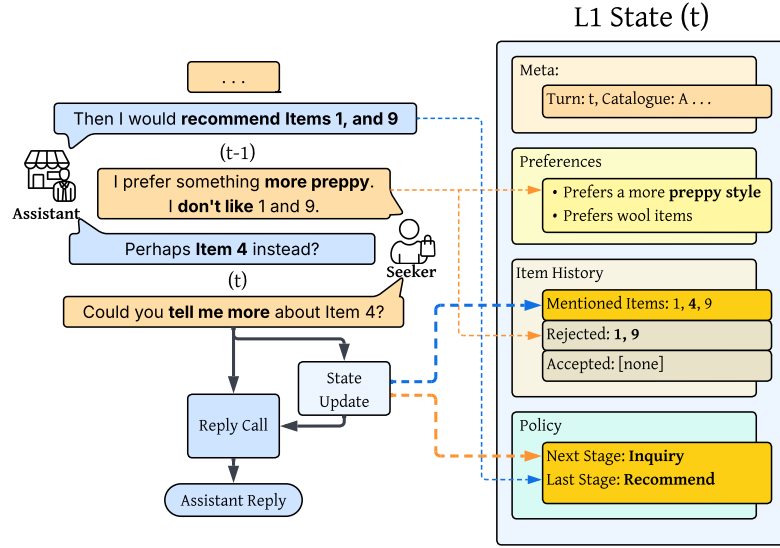


Figure 4.3: Illustrative L1 state update using example conversational content. L1 preserves surfaced preferences, discussed and rejected items, and coarse interaction policy across turns, allowing later responses to remain consistent with earlier feedback.

Tracking [27], and is included here as a representative example of a more standard state-tracking design prior to the fuller IN representation introduced in L2. Fig. 4.3 presents a typical state update step, with conversation-conditioned changes highlighted.

Operationally, L1 inserts a compact state-tracking layer between the raw dialogue history and the final user-facing response. At each turn, the system updates a lightweight dialogue state, then generates the final response conditioned on that updated state together with the transcript and shared catalogue context. The goal is to provide the model with a persistent scaffold for turn-to-turn consistency.

Following Kemper et al. [27], with minor adaptations for our multimodal fixed-catalogue setting, the L1 state is organized around three main components: a *preference state* that records positive, negative, and uncertain signals from the conversation; an *item history* that tracks which catalogue items have been discussed and how they have been received; and a compact *policy state* that represents the current interaction mode at a coarse level, such as requesting additional information, asking a targeted follow-up, or making a recommendation. Following Kemper et al. [27], we use the stages REQ_INFO, INQ, and REC.

L1 is therefore intended as a first design response to the baseline failures, especially to improve turn-to-turn continuity and stage control (W1), follow-up on underspecified preferences (W2), and persistence of feedback across turns (W4).

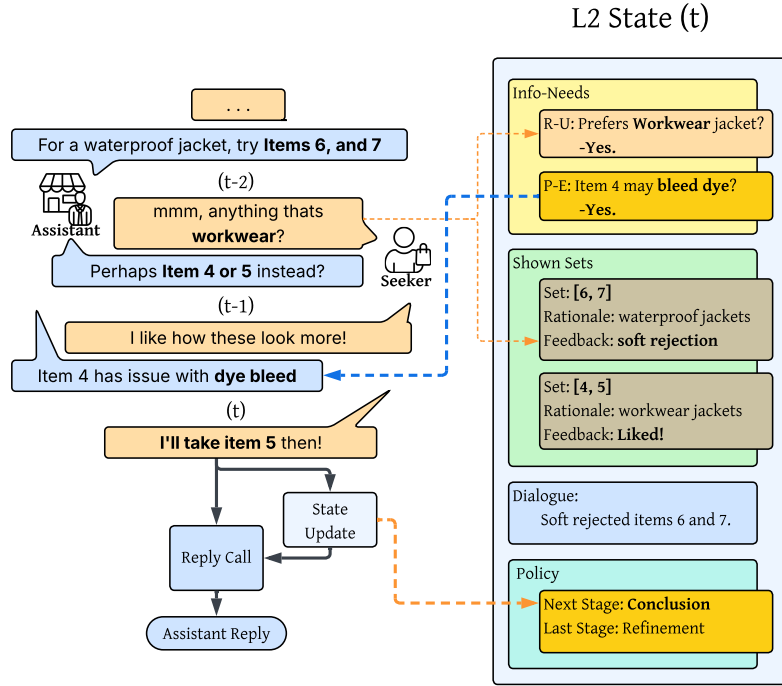


Figure 4.4: Illustrative L2 state update using example conversational content. L2 extends lightweight continuity tracking with unresolved INs, shown-set history, dialogue observations, and richer stage state, allowing the system to better interpret critique and support decision repair across turns.

4.2.3 L2: Information-Needs and Interaction State

L2 is designed as a direct response to the baseline weaknesses that remained unresolved after L1, and thus targets deeper failures involving unresolved criteria, grouped recommendations, critique signals, and decision support.

Conceptually, L2 shifts from a primarily preference-tracking view of conversational recommendation toward a more decision-supportive interaction state. Thus, we aim to preserve both preferences and also decision-relevant information support, along with more intentional parsing of implicit signals and set dynamics. In this sense, L2 serves as the first concrete instantiation of the broader IN perspective developed in this thesis, and of our results from Part I. Fig. 4.4 again presents a state update step, with conversation-conditioned changes highlighted.

The central addition in L2 is an explicit IN structure, following our description in Sec. 3.4.4. Rather than treating uncertainty as generic missing preference information, L2 represents unresolved questions that are still decision-relevant. L2 also adds an explicit parsing of dialogue observations, intended to preserve weak rejection, contrastive feedback within a set, selective enthusiasm, or shifts in emphasis after earlier recommendations. In parallel, the system records shown sets, namely groups of items that were presented together, the rationale under which they were introduced, and any future user feedback.

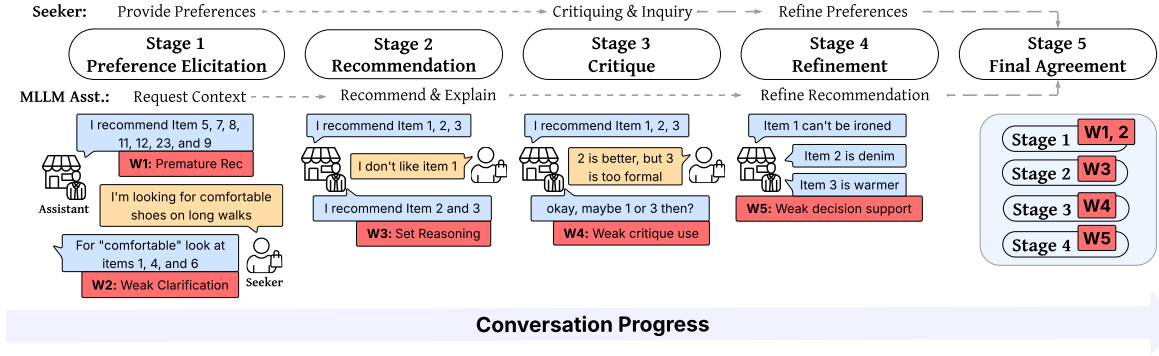


Figure 4.5: Stage-based view of the main baseline weaknesses observed in Part II. The baseline most visibly breaks down during early preference elicitation, however they do not necessarily recover as the conversation progresses.

The policy state is also made more explicitly stage-structured by grounding it in the interaction patterns identified earlier in the VOGUE analysis. In practice, we collapse this into a compact set of policy stages: ELICIT, REC, CRIT, REFINE, and CONCL. This gives the model a more explicit representation of long-horizon or interaction-scale planning, while still allowing flexibility for unpredictable or non-linear conversations in deployment.

Thus, L2 extends L1 by adding explicit unresolved information, dialogue-level observations, shown-set history, and a more explicit stage structure. It is intended as a broader design response to the remaining baseline failures, especially W2, W3, W4, and W5.

4.3 Baseline shortcomings

Live sessions with the baseline revealed recurring interaction failures. The central problem was weak handling of INs over the course of the interaction. Poor IN utilization manifested in several key weaknesses: the system often moved into recommendation before key criteria had been clarified, treated critique weakly once the initial framing proved incomplete, and struggled to turn relevant catalogue evidence into strong support for comparison and final choice. Compared with the staged interaction patterns in VOGUE, the interaction often drifted after early recommendation, even though individual responses usually remained locally coherent and plausible in isolation. This made many failures subtle and easy to overlook in the moment. A key risk, especially in decision-support settings, is that such locally plausible interaction can encourage greater decision confidence than the underlying recommendation process warrants.

For presentation, we organize these shortcomings by the interaction stage in which they appeared most clearly, while recognizing that real conversations often move non-linearly between elicitation, recommendation, critique, and follow-up. Figure 4.5 provides a compact stage-based summary, and Table 4.1 lists the corresponding weaknesses by dominant stage. Transcript excerpts and detailed analysis are available in Appendix B.2.1.

Table 4.1: Baseline weaknesses by dominant interaction stage.

Stage	Code	Failure
S1	W1	The system often recommends too early , before establishing a clear understanding of the user’s needs or what the current turn should accomplish.
	W2	Follow-up questions remain shallow , weakly probing areas of decision uncertainty.
S2	W3	The system struggles to reason over items recommended together as a coherent set with a shared rationale.
S3	W4	User feedback is handled too locally , with weak use of implicit and contrastive feedback.
S4	W5	The system provides weak support for comparison, narrowing, and confident final choice .

4.3.1 Stage 1: Premature recommendation and weak stage control.

W1: Premature recommendation and weak stage control. The baseline often moved into recommendation too early. Rather than spending sustained effort clarifying the user’s goals, contextual constraints, or still-unresolved decision criteria, it frequently collapsed elicitation and recommendation into the same move and introduced candidate items before a stable decision frame had been established. As a result, it surfaced items quickly but invested less in determining what the current turn should accomplish.

W2: Limited clarification of unresolved criteria. A second elicitation problem was shallow probing of underspecified preferences, trade-offs, and scenario-dependent constraints. The system could often ask a plausible opening question, but it less often clarified the deeper decision criterion behind the user’s request or what still needed to be established for the choice. Across sessions, this produced reasonable local follow-up but a relatively shallow diagnosis of what remained uncertain and why it was relevant to the decision.

4.3.2 Stage 2: Recommendation failures

W3: Poor reasoning over recommendation sets. Once recommendation began, the baseline often failed to treat grouped candidates as a coherent set with a shared rationale that could later be refined, narrowed, or abandoned. Feedback on one member of a grouped recommendation did not reliably trigger reconsideration of the recommendation frame as a whole. More broadly, the model treated item mentions too independently, rather than maintaining recommendation sets as structured conversational hypotheses.

4.3.3 Stage 3: Critique and refinement failures

W4: Weak use of implicit and contrastive feedback. After recommendations had been made, the baseline was much more reliable with explicit statements than with weaker conversational signals. Negative or lukewarm feedback did not always produce durable updates, and critique was often treated as a narrow comment about one item rather than as evidence

that the earlier framing had left something unresolved. As a result, the system was less effective at using critique to revise the decision space or identify what still needed clarification.

W5: Grounding without strong decision support. Because the catalogue was small and fully visible in context, the system usually stayed close to the available items and did not often hallucinate unsupported properties. Yet grounding alone did not make the interaction strong. Responses were still repetitive, wordy, and weakly directed, which made it harder to support comparison, reduce uncertainty, and help the user converge on a final choice. This is especially problematic because locally plausible, well-grounded responses can create the appearance of strong support even when the system has not adequately clarified what remains unresolved in the decision or revised its recommendation logic accordingly.

These observations motivate the more structured systems introduced next. L1 addresses the most immediate continuity failures by externalizing lightweight persistent state over user preferences, item history, and coarse interaction policy. L2 extends this by explicitly representing unresolved INs, dialogue-level observations, and recommendation-set history. More broadly, the progression from the baseline to L1 and L2 reflects a shift from minimally structured response generation toward interaction state that more directly represents what remains unresolved in the decision. Both variants are intended to improve interaction quality while reducing the risk of superficially coherent yet weakly supported recommendation trajectories.

4.4 Comparative observations across systems

The baseline shortcomings motivated two successive design responses. L1 was introduced to improve continuity, explicit preference persistence, and coarse stage control. L2 extended this with explicit tracking of unresolved INs, richer dialogue observations, and recommendation-set history. Table 4.2 summarizes the role of each addition, while Table 4.3 reports aggregate outcomes across system conditions.

The two structured variants do not improve in the same way. L1 yields the strongest gains in post-conversation preference alignment, raising PC from 0.560 to 0.699 and reducing MAE from 0.813 to 0.723. L2 remains close on those metrics, with PC of 0.689 and MAE of 0.734, but shows a different interaction profile: it introduces the fewest items of the MLLM variants and uses the fewest turns that bring new items into the conversation. That difference suggests that L1 mainly improves carryover of surfaced preferences, whereas L2 more directly changes how the system handles unresolved criteria during recommendation and refinement.

The qualitative results point in the same direction. L2 does not clearly outperform L1 on the headline alignment metrics. Its gains instead appear most clearly in the recommendation weaknesses identified earlier: via INs, L2 more often keeps the interaction focused on unresolved criteria, follows up better when the user’s real concern is not yet explicit, expands the candidate space less unnecessarily, and reformulates more effectively after critique. These improvements

Table 4.2: Design-response traceability across the Part II system progression. We present changes introduced in L1 and L2, and their observed impacts and shortcomings.

Addition	In:	For:	Observed gains	Gap
Pref. state	L1	W1, W2, W4	Better short-horizon continuity and stronger carryover of already-surfaced preferences , exclusions, and practical constraints across turns.	Improved persistence more than diagnosis; less effective at identifying unresolved criteria not yet made explicit .
Item History	L1	W4	Cleaner local repair after explicit rejection, with less obvious forgetting of discussed or disliked items.	Feedback handling remained largely item-local , with weak revision of the broader recommendation frame.
Policy	L1 & L2	W1, W5	More orderly narrowing and better critique-to-follow-up transitions, with the richer L2 stage structure supporting more selective interaction once earlier framing proved incomplete.	Stage control improved only partially ; real interactions remained non-linear, and gains did not translate into uniformly strong final-stage decision support.
Dialogue obs.	L2	W2, W4	Better use of hesitation, contrast, and selective enthusiasm as evidence that the earlier framing was incomplete.	Gains remained qualitative and inconsistent ; not all implicit or contrastive feedback was handled robustly.
Set-tracking	L2	W3	Some improvement in follow-up after grouped recommendations failed, especially when critique made the missing criterion more explicit.	Recommendation-set reasoning remained only partially improved ; grouped candidates were still often handled too sequentially.
IN-tracking	L2	W2, W5	Stronger follow-up on unresolved criteria and better reformulation when the recommendation frame had missed what still needed to be resolved.	Did not fully solve later-stage decision support or reliably produce stronger final comparison and closure.

are closely tied to the IN-oriented additions in L2, which provide a more explicit basis for representing informational exchange and decision uncertainty. However, some gaps remain. Human Assistants are still much more selective and far less verbose than any of the MLLM variants.

4.5 Design implications

The Part II results support two main design implications. First, live multimodal conversational recommendation benefits from an explicit interaction state. Systems that rely only on the immediate transcript and catalogue context are less reliable at carrying forward surfaced preferences, exclusions, and other decision-relevant signals across turns.

Second, the structure of that state makes a material difference. L1 accounts for most of the gain in post-conversation preference recovery by preserving local preference and policy information more consistently across turns. L2 improves by explicitly representing unresolved INs, more often keeping the interaction focused on decision support and information exchange.

Table 4.3: Aggregate conversational and alignment statistics across system conditions. Fewer introduced items and fewer turns with new items suggest more targeted interaction; higher PC and lower MAE indicate stronger post-conversation preference alignment. Best and second-best results are marked among the three MLLM system variants only.

Metric	Baseline	L1	L2	Human Asst.
Mentioned items ↓	6.6	<u>6.5</u>	5.7	3.55
Turns w/ new items ↓	2.8	<u>2.3</u>	2.2	2.1
Turns to converge ↓	16.5	15.3	15.3	14.8
Conv. Length ↓	17.5	16.3	16.3	13.8
Avg. Asst. Tokens per turn ↓	<u>91.878</u>	89.827	109.913	37.900
PC ↑	0.560	0.699	<u>0.689</u>	0.640
MAE ↓	0.813	0.723	<u>0.734</u>	0.900

Relative to lighter state, L2 stays with unresolved criteria more reliably, handles critique with less drift, and narrows the candidate space with greater discipline. These gains are not mere interaction style change, and instead materially improve recommendation behavior, especially when the initial framing proves incomplete and the system has to recover through clarification and refinement. The alignment gains are smaller, but the recommendation process is better controlled.

At the same time, the remaining limitations are clear. Richer state improved elicitation and critique handling more than it improved set reasoning or final decision support. Feedback on grouped recommendations still could propagate weakly, and comparative evidence was used inconsistently. Explicit IN-oriented state therefore appears useful, but stronger decision support also depends on how that state is structured and how effectively the dialogue policy uses it.

Future work follows directly from these results. Recommendation sets should be treated as first-class interaction objects rather than as loose collections of items. When a user rejects part of a set, the system should be able to revise the rationale behind that set, rather than update only item-level beliefs. Richer IN representations also require stronger dialogue policies. Tracking unresolved INs produced useful gains here, but the representation and policy remain limited, and the system still too often continues eliciting or recommending by default. More broadly, these design choices should be tested in larger and more varied studies to determine how well they generalize across users, scenarios, and domains.

Overall, Part II points to a broader requirement for multimodal conversational recommendation. Better base models and better item grounding are not sufficient on their own. Systems also need an explicit interaction state that represents more than surfaced preferences and that helps determine what still needs to be resolved for the recommendation to move forward.

Chapter 5

Conclusion

Multimodal conversational recommendation relies as much on informational understanding and exchange, as it does preference inference. In realistic settings, users cannot simply be reduced to a set of preferences and constraints to elicit. They are comparing alternatives, identifying decision criteria, and resolving uncertainty before choosing. A system can therefore remain coherent and grounded to explicit user signals while still offering weak decision support.

Part I developed this argument through VOGUE, a human–human multimodal recommendation dataset in fashion. Our analysis shows that these interactions strongly revolve around uncertainty and clarification, with Assistants consciously guiding conversation to enable better user understanding. Preference alignment explains part of conversational success, but a more holistic view is through informational exchange, and how that asymmetry shapes the conversation dynamics. These observations motivate the Information-Needs framework introduced in this thesis.

The framework distinguishes INs by source and temporal status. Some arise directly from the user, while others follow from domain knowledge or catalogue evidence available to the system. Some are already active in the dialogue, while others emerge only after items are presented and compared. Making those differences explicit provides a stronger basis for evaluation and system design, and offers a more needs-based perspective of recommendation as a task.

Part II examined current MLLM systems in a live multimodal recommendation setting. The baseline often produced plausible, catalogue-grounded responses, but those properties were insufficient. A central weakness was poor handling of unresolved INs over the course of the interaction. That weakness appeared in the system’s tendency to recommend too early, handle grouped recommendations weakly, use feedback inconsistently, and provide limited support for comparison and narrowing.

The structured variants show that explicit interaction state improves performance, though the gains are uneven. L1 mainly improved continuity and post-conversation preference recovery. L2 improved recommendation by explicitly representing unresolved INs. That scaffold more

often kept the interaction focused on decision clarification and user understanding, thus reducing drift, and supporting stronger recovery after initial interactions. Weaknesses remained, especially in set-level reasoning, implicit feedback handling, and final decision support.

Taken together, the results indicate that stronger base models and better grounding alone will not produce stronger multimodal conversational recommendation. Systems also need representations and policies that track what remains unresolved for the decision and use that information to guide clarification, comparison, refinement, and closure over time. INs are therefore important in multimodal CRS not only as an analytic framework, but as a specific ability and task which underpins recommendation. Evaluation should reflect that broader objective. Alongside whether a system recommends plausible items, it should assess whether the system sufficiently understands a user, and how well they’ve supported the decision process.

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Appendix A

Part I Supplementary Materials

A.1 Vogue Protocol Supplement

All core protocol materials for VOGUE, including the full scenario texts, survey instruments, and annotation documentation, are available in the anonymous repository.¹ For completeness, and to keep the present document reasonably self-contained, we reproduce the most directly relevant materials here.

A.1.1 Full scenario prompts

1: Choosing the Right Outerwear for a Fall Visit to a Farm.

You're spending the afternoon at a farm in early Fall, picking apples, exploring a corn maze, or visiting a seasonal market. The weather is cool with a chance of light rain, and the ground might be damp or muddy in places. You'll be outdoors for a few hours, walking through grassy fields and uneven dirt paths, bending to pick produce, and possibly carrying bags or baskets. You want your jacket or outerwear to be warm, weather-resistant, and easy to move in, something practical yet put-together enough for photos or a stop at a cafe on the way home.

2: Regular Day on Campus.

It's a typical weekday at university. You've got a few classes, maybe a study group, and you'll be walking between buildings, grabbing coffee, and hanging out around campus. The weather's like the one in late March in Toronto: chilly in the morning, slightly warmer by

afternoon, and there's always a breeze. You want something easy to throw on, comfortable, functional, and ideally something that says a little about your style. What kind of jacket or outerwear would you like to wear for this event?

3: Casual Friday at a Business Casual Office.

It's Friday at a workplace with a slightly cold indoor temperature, where the usual dress code is business casual. Things are more relaxed than usual, but you'll still be around coworkers and may have a few meetings. You want to dress comfortably without looking sloppy, something appropriate for the office, but easygoing enough to carry into after-work plans. Which layering piece would you wear to stay comfortable, warm, and polished in this modern office setting?

4: Dinner Gathering at a Friend's Place (Indoor Social).

It's a weekend evening, and you've been invited to a small dinner at a friend's apartment. It's not formal, no dress code, but you know people will look put together. You'll be indoors, maybe helping in the kitchen, lounging on the couch, or playing a group game. You want to look good without looking like you tried too hard, casual, clean, and comfortable. Which layering piece would you wear in this setting?

5: A Day in Montreal in Winter.

You've set aside a full day during your trip to explore Montreal. You plan to visit markets, museums, and treat yourself to dinner in the evening. You'll be moving between outdoor streets and warm indoor spaces throughout the day, walking on paved sidewalks, transit stations, and store floors. It's winter, but there's no snow on the ground. Which shoes would you like to wear in this case?

6: Warm Day at the Park After Rain.

It's a warm, humid day, and you're heading out for a few hours to explore a large city park, followed by a well-marked trail that leads to scenic viewpoints. It rained earlier, so the ground might still be damp or a little slick in places, especially on grass or dirt paths. You'll be walking a lot and may stop at a cafe or public area along the way. Which shoes would you like to wear in this case?

A.2 Additional analyses for Part I

This section provides supplementary analyses for Part I that support, but do not repeat, the main-body results. We first examine whether the minimum item-discussion instruction meaningfully distorted item-mention behavior. We then provide additional stage-level diagnostics, including dialogue-act transition Sankey diagrams and stage-wise intent histograms.

A.2.1 Minimum item mention check

During data collection, Assistants were instructed to discuss at least three items in each conversation. To verify that this instruction did not mechanically force the observed dialogue structure, we examined the distribution of first item mentions over the course of the conversation for both participant roles.

Figure A.1 shows the cumulative distribution of first item mentions by role. Two patterns are relevant. First, the distributions do not exhibit a visible plateau around two mentioned items, which would be expected if participants were merely satisfying the minimum requirement and then stopping. Second, the final number of mentioned items does not cluster tightly around three. Instead, the observed distribution is more consistent with the broader interaction pattern discussed in the main body, namely that Assistants often introduce an initial cluster of roughly two to four items and then add further candidates only if needed. The modal number of Assistant first-mentioned items is four. Taken together, these patterns suggest that the minimum-item instruction did not materially determine the dialogue structure.

A.2.2 Additional stage-level diagnostics

The main body introduces a five-stage view of VOGUE conversations and uses it to interpret the broad dialogue structure. Here, we provide additional diagnostics that make the stage patterns more explicit, without repeating the main narrative description. In particular, we include stage-wise Sankey diagrams of dialogue-act transitions (Fig. A.2) and stage-wise histograms of dialogue-act distributions (Fig. A.3).

These supplementary plots reinforce three aspects of the stage analysis. First, the early-stage patterns in Fig. A.2 and Fig. A.3 are more asymmetric, with Assistants more often initiating task progress through requests and recommendation moves while Seekers primarily provide context, preferences, and early reactions. Second, the middle stages in both figures show stronger back-and-forth structure, especially around critique, inquiry, comparison, and explanation. Third, the final-stage distributions, particularly in Fig. A.3, concentrate around acceptance and closure-oriented acts rather than continued broad exploration. Although Seekers were permitted to reject all options, this did not occur in the collected conversations.

Table A.1: Seeker intent, building on [22, 23]. We present them hierarchically, with higher-fidelity tags grouped under broader categories. Tags marked with [†] are newly introduced in VOGUE.

Category (Code)	Description (Seeker...)	Example
Ask for Recommendation		
Initial Query (IQ)	...asks for a recommendation in the first query	“Hey, I want to purchase one coat...”
Continue (CON)	...asks for another recommendation	“Would you give me a Plan B instead?”
Provide		
Provide Context (PCT)	...provides background or situation information	“I want to choose outerwear for a fall visit to a farm...”
Provide Explicit Preferences (PEP) [†]	...states explicit preferences, intent, constraints	“It has to be waterproof.”
Provide Implicit Preference (PIP) [†]	...reveals preferences indirectly	“I don’t really like Item11 compared to Item10.”
Refine Preference (RP)	...refines or relaxes previously stated preferences	“I don’t need a hood necessarily...”
Answer (ANS)	...answers a question from the Assistant	“The weather is cool with light rain.”
Acknowledgement (ACK)	...shows understanding or confirmation	“Yeah.”
Recommendation Rating		
Interest (INT) [†]	...reacts positively without a final choice	“I do like Item08 and Item05.”
Accept (ACT)	...accepts the recommended item	“I think I’ll go with Item02.”
Reject (RJT)	...rejects the recommendation	“Item03 is too ugly for me.”
Neutral Response (NR)	...does not indicate acceptance or rejection	“I see.”
Inquire		
Inquire – Factual (IF) [†]	...asks for factual information about an item	“Does it have a pocket?”
Inquire – Ask Opinion (IA) [†]	...asks for the Assistant’s subjective opinion	“Which one do you think is more comfortable?”
Critiquing		
Critique – Feature (CF)	...critiques a specific attribute of an item	“Item05 is a bit too long.”
Critique – Compare (CC)	...requests comparison between items	“Between Item23 and Item16, which is better?”
Explain (EXP)[†]	...explains reasons behind verdict or preference	“I like Item05 because it goes well with my outfit.”
Ask Clarification (AC)	...asks clarification of something previously said	“Did you say Item13 as well?”
Others (OTH)	Greetings, gratitude, or chit-chat	“Hello.”

Table A.2: Assistant intent, building on [22, 23]. We present them hierarchically, with higher-fidelity tags grouped under broader categories. Tags marked with [†] are newly introduced in VOGUE.

Category (Code)	Description (Assistant...)	Example
Request		
Request Task Initiation (RTI) [†]	...initiates dialogue with an open-ended prompt	“How can I help you today?”
Request Preferences (RP) [†]	...elicits the Seeker’s preferences or constraints	“Are you looking for something longer?”
Request Context (RC) [†]	...elicits situational context	“Where is this farm going to be?”
Clarify Question (CQ)	...asks clarification of a prior requirement	“Do you mean a more formal setting?”
Ask Opinion (A)	...asks the Seeker to choose or judge among options	“Which one do you prefer?”
Ensure Fulfillment (EF)	...checks whether the task is complete	“Any other items you want to see?”
Inform Progress (IP)	...indicates processing or navigation state	“Give me one second, I’m checking Item29...”
Acknowledgement (ACK)	...acknowledges a Seeker utterance	“Got it.”
Answer (ANS)	...answers a Seeker’s question	“Waterproof is stronger than water-resistant.”
Recommend		
Recommend – Show (RS)	...presents candidate items	“Take a look at Item31, Item32, and Item28.”
Recommend – Combine (REC) [†]	...recommends compatible combinations or layered outfits	“You can wear Item22 under Item17.”
Explain		
Explain – Preference (EP)	...justifies recommendations using preferences	“Item03 is best because it’s waterproof.”
Explain – Additional Info (EAI)	...adds new factual or descriptive details	“It’s machine washable.”
Personal Opinion		
Comparison (PCM)	...compares two or more items	“Only Item11 is machine washable.”
Persuasion (PER)	...adds positive framing or encouragement	“Both materials are pretty good.”
Prior Experience (PEX)	...references personal experience	“I have one myself, it’s my go-to boot.”
Context (PCN)	...gives opinion grounded in context	“This matters more in deep snow.”
Others (OTH)	Greetings, gratitude, or chit-chat	“Hi.”

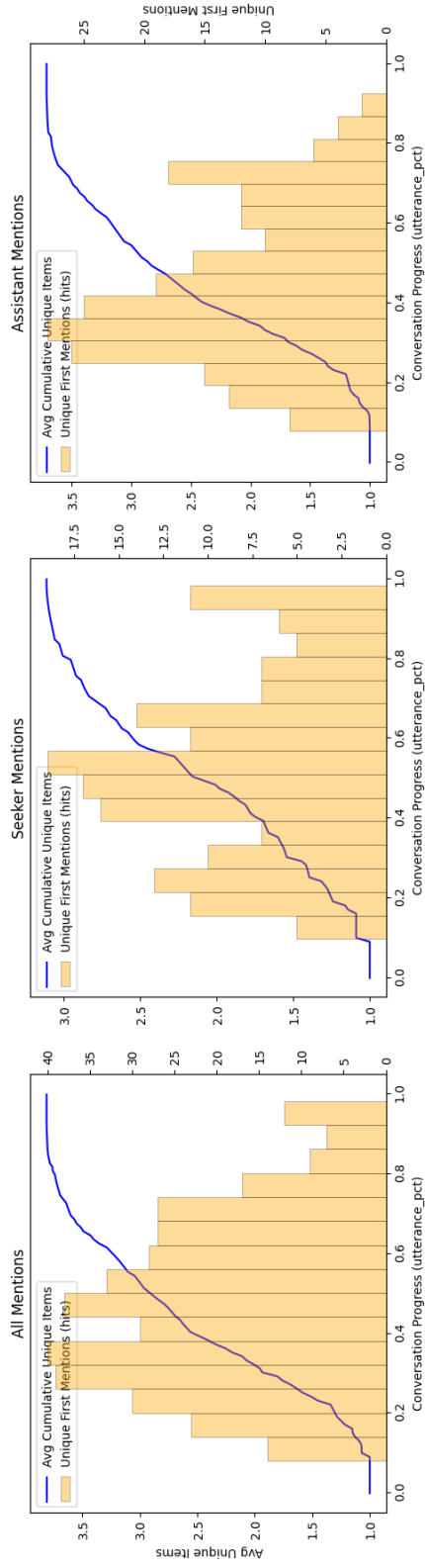


Figure A.1: Cumulative distribution of first item mentions over the course of the conversation, shown separately by participant role. The lack of a visible plateau around two items, and the absence of strong clustering around the instructed minimum of three, suggest that the minimum-item instruction did not mechanically determine the observed dialogue structure.

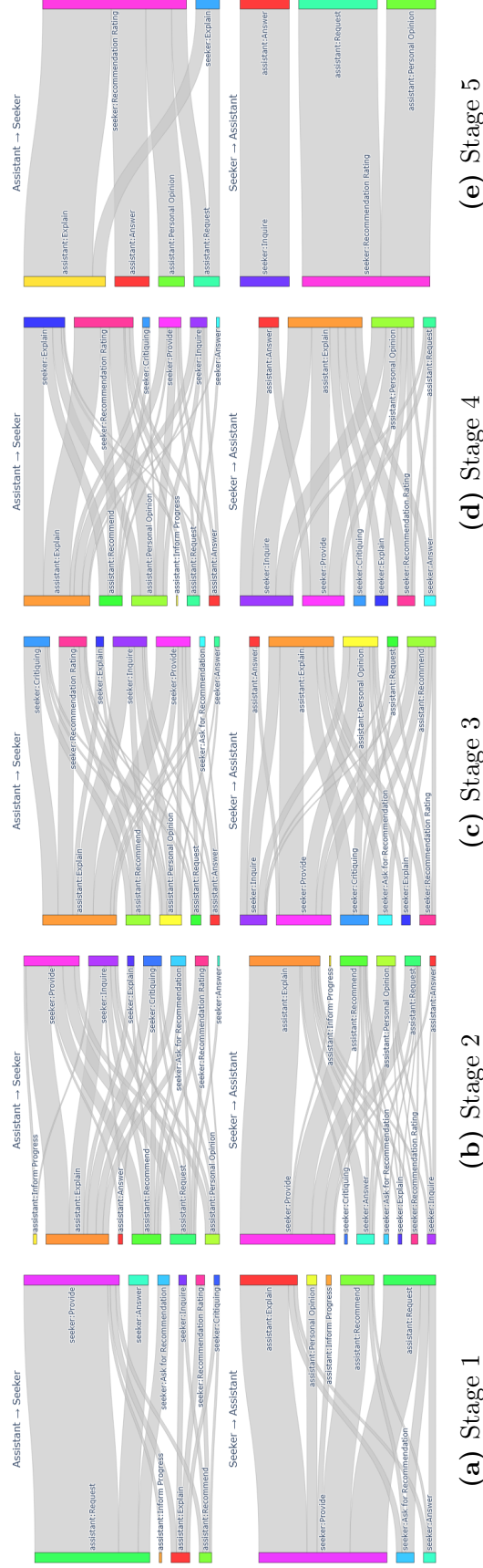


Figure A.2: Stage-wise Sankey diagrams of dialogue-act transitions. These diagrams provide a more detailed view of how interactional flow changes across the five stages identified in the main text.

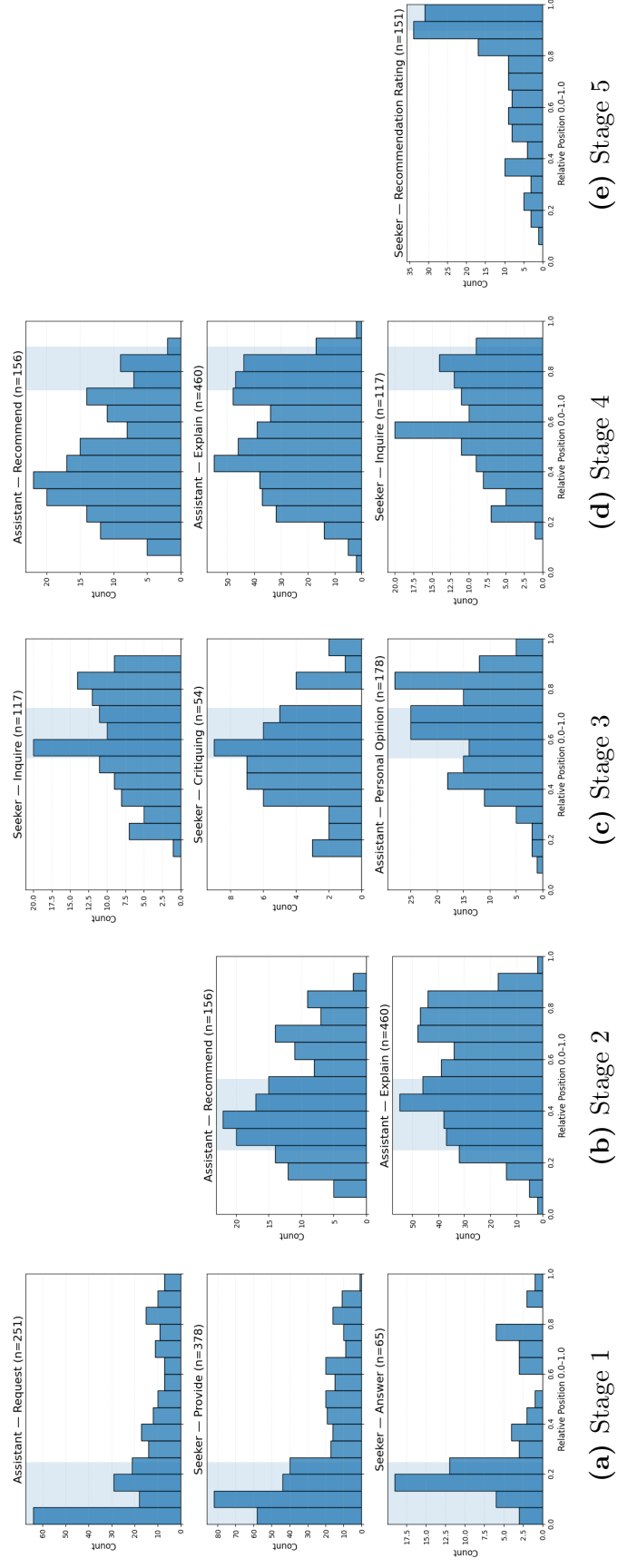


Figure A.3: Stage-wise histograms of dialogue-act distributions. These plots show the relative concentration of intent categories within each stage.

Appendix B

Part II Supplementary Materials

This appendix provides supplementary material for Part II. We first present additional methodological details for the Part II test bench, system prompts, and implementation choices. We then provide additional qualitative analysis of the baseline and structured systems, organized around the main interaction patterns discussed in Chapter 4.4.

B.1 Methods for Part II

This section provides additional methodological detail for the Part II test bench beyond what is included in the main chapter. Prompts, state schema, update and reply contracts are included verbatim.

B.1.1 Per-turn call structure and provided context

The three Part II conditions differed in whether they maintained explicit interaction state, how many model calls they made per turn, and what context each call received. In all cases, the server maintained the running transcript and reconstructed the active 12-item catalogue context at each turn. In the shipped presets, the model also had access to the full catalogue metadata each turn and a composite catalogue grid image was attached as visual context.

The **baseline** used a single model call per turn. This call was the final user-facing reply generation step. It was given the current user message, the running transcript, the full catalogue metadata, the composite catalogue image, and any lightweight session context (e.g. turn number, catalog number) maintained by the server. It did not include a separate planner or model-side state-update phase. The baseline reply prompt therefore had to handle interpretation of the latest turn and production of the next response in a single pass.

L1 used a two-step turn structure. First, a planner/state-update call received the running transcript, the current L1 state, the catalogue metadata and image context, and the L1 schema and update-contract instructions; its role was to return an update patch rather than a user-facing

reply. Second, a final reply-generation call received the transcript, the updated L1 state, the catalogue metadata and image context, and the L1 schema and reply contract. The L1 system therefore separated interpretation and state maintenance from user-facing response generation, while grounding the reply primarily in compact preference, item-memory, and stage information.

L2 used the same overall three-step structure as L1, but with a richer state representation and correspondingly richer prompt contracts. Its planner/state-update call received the running transcript, the current L2 state, the full catalogue context, and the L2 schema and update-contract instructions. Its final reply-generation call then used the updated L2 state together with the transcript and catalogue context, grounding the reply not only in coarse stage and item history, but also in INs, dialogue observations, and shown recommendation sets. Relative to L1, the L2 calls therefore had access to a broader interaction representation oriented toward unresolved criteria, critique, and set-aware continuity.

Across all three systems, the final reply stage enforced several shared behavioral constraints. In particular, replies were required to refer to products only as `item <id>`, to avoid unsupported claims, and to ignore price, size, and color in both reasoning and response generation. In L1 and L2, the final reply additionally appended a compact `ASSISTANT_OUTCOME` object so that newly introduced items or item sets could be recorded by the server for later turns.

B.1.2 Prompt and contract texts

This subsection reproduces the main prompt and contract artifacts used by the deployed Part II conditions. In the shipped runtime, the baseline used a single final-reply prompt per turn, while L1 and L2 added a planner/state-update pass together with explicit schema, update, and reply contracts. L1 used a lightweight preference-and-memory state, whereas L2 extended this with INs, dialogue observations, and shown-set tracking.

Shared final-reply prompt

The baseline uses this prompt directly. In L1 and L2, the same reply-generation prompt is used together with the relevant schema reference and reply contract; in the structured settings it also appends an `ASSISTANT_OUTCOME` object after the natural-language reply to enable some server-side logistics and state management.

```
=====
You are generating the next user-facing reply for a conversational
recommendation assistant.
Your task is to produce the best next assistant reply for the current
turn.
```

Prioritize the latest user message, then use any available supporting context to make the reply coherent, grounded, and useful.

Write a natural, concise (about 1-6 sentence), user-facing reply that moves the conversation forward.

Be helpful without sounding repetitive or over-explaining.

Prefer one clear next step over several loosely related ones.

=====

Use the provided state and transcript, along with catalogue information.

=====

Do NOT expose internal field names, state structure, policy labels, or system machinery.

=====

Reference items only as 'item <id>'.

Do not use product titles or names.

Do not restate already-surfaced item attributes unless clarification is necessary.

Do not make unsupported claims.

Surface uncertainty when needed.

=====

DO NOT USE PRICE, SIZE, or COLOR preferences when reasoning or replying. These should be ignored and assumed irrelevant.

=====

If the user indicates they've made their final selection, you can assume the conversation over, do not discuss sizing, shipping, pickup, or actual purchase logistics.

Shared planner/state-update prompt (L1 & L2)

=====

You are performing the planner/state-update pass for a conversational recommendation assistant.

Your job is to interpret the latest turn and update the conversation state for server-side merge.

Inputs may include:

- the conversation transcript
- the current state
- tool outputs, if any
- catalogue metadata and images, if any

Your responsibilities:

- update the state based on the latest turn
- preserve prior information unless it is explicitly negated, replaced, rejected, or resolved
- keep updates grounded in the transcript and available evidence
- keep outputs concise and structured

Do NOT:

- answer the user
- draft or script the final assistant reply
- explain your reasoning
- output prose outside the state update tool c

=====

Domain constraint

Ignore PRICE, SIZE, and COLOR when reasoning about:

- constraints
- preferences
- plans
- assistant questions

Do not add them to state.

Do not propose questions about them.

=====

Follow the attached schema note for:

- writable fields
- always-refresh fields
- patch-only fields
- server-owned fields
- clearing rules
- section-specific semantics

Keep info_needs focused on decision-relevant user/expert requirements.

If an info need can be answered this turn, provide the answer and move status to DISCLOSED or RESOLVED.

=====

Return exactly the update required by the schema note and call update_state once.

L1 state template

```
{
  "meta": {
    "schema_version": "v2",
    "turn_index": 0,
    "catalog_variant": "<catalogue letter/name>",
    "session_goal_nl": "<e.g. mission summary, find winter coat>"
  },
  "preference_state": {
    "constraints": [
      {
        "slot": "<e.g. budget>",
        "value": "<e.g. <=150>",
        "confidence": 0.8
      }
    ],
    "preferences": [
      {
        "description": "<e.g. user prefers long sleeves over t-shirts>",
        "confidence": 0.7
      }
    ]
  },
  "item_memory": {
    "mentioned_item_ids": [
      ""
    ],
    "rejected_item_ids": [
      "<e.g. 4>"
    ]
  }
}
```

```

    ],
    "recommended_item_ids": [
        "<e.g. 9>"
    ],
    "accepted_item_ids": [
        "<e.g. 7>"
    ]
},
"policy": {
    "next_stage": "<UNK>",
    "last_stage": ""
},
"schema_reference_file": "l1_schema_reference.md",
"update_contract_file": "l1_update_contract.md",
"reply_contract_file": "l1_reply_contract.md"
}

```

L1 schema reference

```

=====
## Sections

### meta
- 'schema_version': schema identifier for the state format.
    Server-controlled.
- 'turn_index': current conversation turn number. Server-controlled.
- 'catalog_variant': identifier for the active catalogue variant.
    Server-controlled.
- 'session_goal_nl': short natural-language summary of the user's
    shopping mission.

### policy
- 'next_stage': the assistant's next action mode. One of:
    - 'REQ_INFO'
    - 'INQ'
    - 'REC'

```

- 'last_stage': previous action mode. Server-controlled.

Stage meanings:

- 'REQ_INFO': essential information is still missing, so the assistant should ask for it before recommending.
- 'INQ': the user is primarily asking about already-discussed items, options, or attributes.
- 'REC': there is enough information to recommend or continue recommending items.

preference_state

- 'constraints': hard requirements or must-haves.
 - Each entry has:
 - 'slot': the constrained attribute/category
 - 'value': the required value
 - 'confidence': confidence score in '[0,1]'
- 'preferences': soft likes, dislikes, or tendencies.
 - Each entry has:
 - 'description': concise natural-language preference
 - 'confidence': confidence score in '[0,1]'

Interpretation rules:

- Constraints are stronger than preferences.
- Do not add weak guesses unless supported by the conversation.

item_memory

- 'mentioned_item_ids': items surfaced or discussed in the conversation.
- 'rejected_item_ids': items the user explicitly declined or ruled out.
- 'recommended_item_ids': items the assistant recommended.
- 'accepted_item_ids': items the user accepted, shortlisted, or positively selected.

Interpretation rules:

- Do not assume acceptance from mild interest unless the conversation clearly supports it.

Invariants

- Server-controlled fields must not be model-written in update mode.
- Item memory fields contain item ids only.
- Unused sections for L1 should not be invented.

L1 update contract

```
=====
## Writable sections
The model may write only:
- 'meta.session_goal_nl'
- 'policy.next_stage'
- 'preference_state'
- 'item_memory'
=====

## Server-owned fields
Do NOT write:
- 'meta.schema_version'
- 'meta.turn_index'
- 'meta.catalog_variant'
- 'policy.last_stage'
=====

## Always-refresh every turn
You MUST include these on every turn (no exceptions):
- 'policy.next_stage'
If any of the above is omitted, the update is invalid.
These fields must be present every turn, even if some values remain
    unchanged.
=====

## Patch-only sections
Include only if changed:
- 'meta.session_goal_nl'
- 'preference_state'
- 'item_memory'

If unchanged, omit them.
=====
```

```

## Clearing rule
Do NOT clear, empty, null out, or weaken a populated field unless the
  conversation explicitly:
- negates it
- replaces it
- rejects it
- resolves it
=====
## Section semantics

### meta
- 'session_goal_nl': update only when the shopping mission is newly
  established, clarified, or changed.

### policy
- MUST be updated every turn to inform what the next assistant reply
  should accomplish.

Priority rule:
- If essential information is still missing, choose 'REQ_INFO'.
- Else if the user is mainly making an inquiry, choose 'INQ'.
- Else choose 'REC'.

### preference_state
Update when preferences/constraints change, or new ones surfaced.
- 'constraints': hard requirements
- 'preferences': soft likes/dislikes
- each entry includes 'confidence' in '[0,1]'

### item_memory
Only update when changed.

Update rules:
- do not invent item ids not grounded in the current state, transcript,
  or tool outputs
=====
## Output contract

```

Call 'update_state' with:

- all always-refresh fields
- any changed patch-only sections

Do NOT output a full-state rewrite.

Do NOT output unchanged patch-only sections.

Only include item ids, never titles.

Keep lists concise.

L1 reply contract

```
=====
## Core behavior
Use:
- 'policy.next_stage'
- 'meta.session_goal_nl'
- 'preference_state'
- 'item_memory'

as the primary grounding for the reply.
=====
## Stage behavior

### REQ_INFO
- Ask for the most important missing information needed to make a better
  recommendation.
- Keep the question focused and minimal.
- Do not ask multiple unrelated questions at once unless necessary.

### INQ
- Answer the user's inquiry about already-discussed items, options, or
  attributes.
- Ground the answer in the known state and available evidence.
- Do not contradict prior item memory.

### REC
- Recommend or continue recommending items.
```

```

- Use known constraints and preferences.
- Avoid recommending items already clearly rejected unless the user
  explicitly reopens them.
=====
## Memory usage
- Respect 'rejected_item_ids'.
- Respect 'accepted_item_ids'.
- Use 'mentioned_item_ids' and 'recommended_item_ids' to maintain
  continuity and avoid repetition.
=====
## Style/behavior constraints
- Do not mention internal stage names.
- Do not mention internal field names.
- Do not describe the state object.
- Keep the reply natural and user-facing.

```

L2 state template

```

{
  "meta": {
    "schema_version": "v2",
    "turn_index": 0,
    "catalog_variant": "<catalogue letter/name>",
    "session_goal_nl": "<e.g. mission summary, find winter coat>"
  },
  "policy": {
    "next_stage": "<UNK>",
    "next_stage_plan_nl": "<e.g. ask about fit preference before showing
next items>",
    "last_stage": "",
    "last_stage_plan_nl": ""
  },
  "info_needs": {
    "surfaced_user": [
      {

```

```

        "question": "<e.g. Does either candidate violate the user's
cruelty-free constraint?>",
        "answer": "<unanswered, answer me if possible>",
        "status": "OPEN"
    }
],
"surfaced_expert": [
    {
        "question": "<e.g. infer: 'Which candidate is more comfortable
for prolonged walking?'>",
        "answer": "<unanswered, answer me if possible>",
        "status": "OPEN",
        "evidence": "<e.g. The user says they will be walking all day at
an outdoor event, but never mentions comfort directly.>"
    }
],
"proactive_user": [
    {
        "question": "<e.g. Does the current candidate satisfy the user's
likely material preference?>",
        "answer": "<proactively surface answer>",
        "status": "OPEN",
        "evidence": "<The user has repeatedly rected items with certain
materials, but has not asked about the current candidate.>"
    }
],
"proactive_expert": [
    {
        "question": "<e.g. Does this item have a known dye-transfer
issue?>",
        "answer": "<proactively surface answer>",
        "status": "OPEN",
        "evidence": "<Reviews mention dye issue, but user would not know
to ask, we should disclose this>"
    }
]
},

```

```

"dialogue_observations": {
  "implicit_signals": [
    "<e.g. ignored item 4 -> NOT_ENDORSED>"
  ],
  "feedback_summary_nl": "<e.g. liked item 9, neutral on others>"
},
"item_memory": {
  "mentioned_item_ids": [
    ""
  ],
  "rejected_item_ids": [
    "<e.g. 4>"
  ],
  "shown_sets": [
    {
      "item_ids": [
        "<e.g. 4>",
        "<e.g. 9>"
      ],
      "assistant_rationale_nl": "<e.g. dark palette long silhouette>",
      "user_response_nl": "<e.g. liked 9>"
    }
  ],
},
"schema_reference_file": "l2_schema_reference.md",
"update_contract_file": "l2_update_contract.md",
"reply_contract_file": "l2_reply_contract.md"
}

```

L2 schema reference

//

```

=====
## Sections

### meta

```

- 'schema_version': schema identifier for the state format.
Server-controlled.
- 'turn_index': current conversation turn number. Server-controlled.
- 'catalog_variant': identifier for the active catalogue variant.
Server-controlled.
- 'session_goal_nl': short natural-language summary of the user's shopping mission.

policy

- 'next_stage': the assistant's next action mode. One of:
 - 'ELICIT'
 - 'REC'
 - 'CRIT'
 - 'FUP'
 - 'CONCL'
 - 'OTH'
- 'next_stage_plan_nl': brief natural-language description of the immediate conversational objective for the current turn.
- 'last_stage': previous action mode. Server-controlled.
- 'last_stage_plan_nl': previous plan description. Server-controlled.

Stage meanings:

- 'ELICIT': important non-excluded preference information is still missing or unclear.
- 'REC': there is enough information to recommend or continue recommending.
- 'CRIT': the user is reacting to shown items and evaluation, comparison, or critique is needed.
- 'FUP': the user is asking a follow-up, clarification, or logistical question about already-discussed items.
- 'CONCL': the interaction should be wrapped up with a concise conclusion and clear closure.
- 'OTH': OTHER, no stage suitable, describe what to do in 'next_stage_plan_nl'

info_needs

- ‘surfaced_user’: decision-relevant needs already ASKED/ANSWERED in dialogue and grounded in user goals, constraints, comparisons, preferences, or dialogue signals. These are analogous to recording user statements, claims, requests etc. If the user asks an explicit question, it goes HERE.
- ‘surfaced_expert’: implicit decision-relevant needs which we infer from the conversation thus far, only if they have not been explicitly surfaced. e.g. The user says they will be walking all day at an outdoor event, but never mentions comfort directly.
- ‘proactive_user’: likely future user-side info needs not yet active, inferred from user goals, behaviors, or actions. The answer would likely help the user if disclosed. We aim to proactively disclose the answer.
- ‘proactive_expert’: expert-side needs who’s answer is worth proactively surfacing because domain knowledge/catalogue evidence suggests choice impact and which the user would likely not know to ask about. We aim to proactively disclose the answer.

Each entry contains:

- ‘question’: the info need, formulated as a question whose answer is decision-relevant.
- ‘answer’: current best answer to the need. Use ‘<unanswered>’ when still unresolved.
- ‘status’: either:
 - ‘OPEN’
 - ‘DISCLOSED’
 - ‘RESOLVED’
- ‘evidence’: evidence and guidance regarding the need.

Interpretation rules:

- Information needs are decision-relevant question-answer requirements whose answer would help the User reach a final decision. Some needs require we disclose the QUESTION to gather an ANSWER, some require that we proactively DISCLOSE the ANSWER early.
- All needs are formulated as a question, whose answer is decision relevant, they are not literal questions.
- User-side needs are not limited to literal user questions.

- Preserve unresolved entries unless the conversation justifies resolving them.
- Keep the lists concise and useful.
- If a need can be answered from explicit user input, grounded inference, or known/disclosed evidence, fill 'answer' and set status to 'DISCLOSED' or 'RESOLVED'.

dialogue_observations

- 'feedback_summary_nl': one-line summary of the latest user reaction, feedback, intent, or shift.
- 'implicit_signals': concise inferred signals that are newly supported by the conversation.

item_memory

- 'mentioned_item_ids': items surfaced or discussed in the conversation.
- 'rejected_item_ids': items the user explicitly declined or ruled out.
- 'shown_sets': historical item sets previously surfaced together.

Each 'shown_sets' entry contains:

- 'item_ids': ids in that set
- 'assistant_rationale_nl': brief reason the set was surfaced
- 'user_response_nl': brief summary of the user's reaction to that set

Interpretation rules:

- Use item ids only, never item titles.
- 'item_ids' are immutable once created.
- Historical set memory is used for continuity, comparison, and critique.

=====

Invariants

- Server-controlled fields must not be model-written in update mode.
- Item memory fields contain item ids only.
- 'shown_sets.item_ids' are immutable once established.

L2 update contract

```
=====
## Writable sections
The model may write only:
- 'meta.session_goal_nl'
- 'policy.next_stage'
- 'policy.next_stage_plan_nl'
- 'info_needs'
- 'dialogue_observations.feedback_summary_nl'
- 'dialogue_observations.implicit_signals'
- 'item_memory'
=====

## Server-owned fields
Do NOT write:
- 'meta.schema_version'
- 'meta.turn_index'
- 'meta.catalog_variant'
- 'policy.last_stage'
- 'policy.last_stage_plan_nl'
=====

## Always-refresh every turn
You MUST include these on every turn (no exceptions):
- 'policy.next_stage'
- 'policy.next_stage_plan_nl'
- 'dialogue_observations.feedback_summary_nl'
- full 'info_needs' object:
  - 'surfaced_user'
  - 'surfaced_expert'
  - 'proactive_user'
  - 'proactive_expert'

If any of the above is omitted, the update is invalid.

These fields must be present every turn, even if some values remain
  unchanged.
=====
```

```

## Patch-only sections
Include only if changed:
- 'meta.session_goal_nl'
- 'dialogue_observations.implicit_signals'
- 'item_memory'

If unchanged, omit them.
=====

## Clearing rule
Do NOT clear, empty, null out, or weaken a populated field unless the
    conversation explicitly:
- negates it
- replaces it
- rejects it
- resolves it
=====

## Section semantics

### meta
- 'session_goal_nl': update only when the shopping mission is newly
    established, clarified, or changed.

### policy
- THESE FIELDS MUST BE UPDATED to inform what the next assistant reply
    should accomplish.
- 'next_stage' must be one of: 'ELICIT', 'REC', 'CRIT', 'FUP', 'CONCL',
    'OTH'.
- 'next_stage_plan_nl': which stage to move to for next turn.
- 'next_stage_plan_nl': brief statement of the immediate conversational
    objective, list any specific questions, task, items, that should be
    accomplished.

### info_needs
Always return the full current object:
- 'surfaced_user'
- 'surfaced_expert'
- 'proactive_user'

```

- 'proactive_expert'

Each entry must include:

- 'question': the info need, formulated as a question whose answer is decision-relevant
- 'answer': current best answer. Use '<unanswered, answer me if possible>' when unresolved.
- 'status': either:
 - 'OPEN'
 - 'DISCLOSED'
 - 'RESOLVED'
- 'evidence': If a need is explicit, give evidence. If a need is inferred, indicate how you made the inference. Indicate whether to disclose the ANSWER or the QUESTION.

Status rules:

UPDATE ALL NEEDS WITH THEIR MOST CURRENT STATUS

- Use 'OPEN' only for unresolved items.
- Use 'DISCLOSED' once the need has been disclosed/brought up in conversation.
- Set 'RESOLVED' if the need has been addressed in a prior turn.

Preserve unresolved entries unless the conversation justifies resolving them.

Avoid speculative clutter.

dialogue_observations

- MUST refresh every turn. 1-2 sentences each.
- 'implicit_signals' any implicit signals, e.g. didn't like item 4, but ignored 2 and 3, which might mean they're also not great.
- 'feedback_summary_nl' must summarize the most recent user utterance (current turn).

item_memory

Only update when changed.

```

Allowed fields:
- 'mentioned_item_ids'
- 'rejected_item_ids'
- 'shown_sets'

For 'shown_sets':
- do NOT add new entries, this is server-owned
- do NOT modify 'item_ids'
- only update:
  - 'assistant_rationale_nl'
  - 'user_response_nl'
- Your update should be grounded in any feedback given by the user for
  that set in the latest turn, likes, dislikes, etc.
- The relevant shown-sets MUST receive a fresh 'user_response_nl'
  grounded in the current turn.
=====
## Output contract
Call 'update_state' with:
- all always-refresh fields
- any changed patch-only sections

Do NOT output a full-state rewrite.

Do NOT output unchanged patch-only sections.

Only include item ids, never titles.

Keep lists concise.

```

L2 reply contract

```

=====
## Core behavior
Use:
- 'policy.next_stage'
- 'policy.next_stage_plan_nl'

```

```

- 'meta.session_goal_nl'
- 'info_needs'
- 'dialogue_observations'
- 'item_memory'

as the primary grounding for the reply.
=====

## Stage behavior

### ELICIT
- Elicit information that advances the conversation toward a better
  recommendation.
- Avoid broad, numerous, or redundant questioning.

### REC
- Recommend or continue recommending using current memory and plan.
- Maintain continuity with previously discussed items and user reactions.
- Avoid resurfacing clearly rejected items unless the user reopens them.

### CRIT
- Compare, critique, or evaluate shown items using remembered sets and
  user feedback.
- Use 'shown_sets' and 'feedback_summary_nl' to ground the comparison.

### FUP
- Answer the follow-up directly.
- Use relevant 'info_needs' buckets ('surfaced_user', 'surfaced_expert',
  'proactive_user', 'proactive_expert'), item memory, and prior
  dialogue context as needed.

### CONCL
- Conclude the interaction clearly and concisely.
- Summarize the final recommendation or outcome without introducing new
  items unless explicitly requested.
- End with closure-oriented language (for example, an invitation to
  return if they want more help).

```

```

### info_needs
- if a need is immediately relevant this turn, and addressing it would
  further the conversational goal, address it. Do not overwhelm the
  user, pick the highest impact needs. To address a need, look to the
  need's 'evidence' field for guidance.
=====

## Memory usage
- Respect 'rejected_item_ids'.
- Use 'mentioned_item_ids' and 'shown_sets' for continuity.
=====

## Style/behavior constraints
- Do not mention internal stage names.
- Do not mention internal field names.
- Do not describe the state object.
- Keep the reply natural and user-facing.
- Do not ask about 'PRICE', 'SIZE', or 'COLOR'.

```

B.2 Additional analyses for Part II

This appendix expands the qualitative analysis for Part II. We first detail the baseline failure modes by interaction stage, then compare how L1 and L2 changed those patterns. We close with a brief synthesis of the main gains and remaining limitations.

B.2.1 Baseline qualitative failure modes

The baseline failures extended beyond isolated item mismatches and often reflected weak handling of unresolved INs over the course of the interaction. This appeared across early elicitation, grouped recommendation, and critique-driven refinement, where the system often failed to identify what still needed to be clarified, compared, or revised for the decision. We therefore organize the failures by the stage where they appeared most clearly, using the staged structure identified earlier in VOGUE.

Stage 1: Elicitation failures

W1: Premature recommendation and weak stage control. The baseline often moved into recommendation too early. Rather than clarifying the user's goals, contextual constraints, or decision criteria, it quickly proposed candidates and continued recommending

until the user either accepted an item or explicitly forced a correction. One early exchange illustrates the pattern:

User: “hi, looking for some good shoes for walking around, like sightseeing shoes”

Assistant: “For lots of walking and sightseeing, I’d focus on comfy, supportive sneakers or light hiking shoes rather than boots or loafers.

From this set, item 25, item 26, item 28, item 31, item 32, and item 35 all look like good candidates since they emphasize cushioning and support.

To narrow it down: do you expect mostly city streets, some light trails/uneven paths, or more rugged outdoor terrain?”

Here the system introduces six of twelve items at once while combining elicitation and recommendation in a single turn. In *VOGUE*, conversations more often progressed through a clearer sequence of elicitation, recommendation, critique, refinement, and agreement. The baseline flattened that progression, surfacing candidates quickly but investing less in establishing what the turn should accomplish.

W2: Limited clarification of unresolved criteria. A second elicitation problem was weak probing of underspecified preferences, trade-offs, and scenario-dependent constraints. The baseline could ask a plausible opening question, but it often failed to clarify the deeper decision criterion behind the request. This was especially visible when the user’s description was broad but still decision-relevant. For example:

User: “I have to go to work and the dress code is usually business casual, but I want something comfortable inside to wear”

Assistant: “For a business-casual office and all-day comfort, I’d focus on soft knits that look neat but feel relaxed.

From what you’re seeing here, item 13 or item 15 would work well, while item 18 and item 19 give you cardigan options. Item 22 is a more fitted turtleneck.

Are you leaning more toward a classic crewneck, a cardigan you can unbutton, or something like a turtleneck?”

User: “I want something like a zip up or something I can easily take off”

Assistant: “For something you can unzip and shrug off easily, item 18 or item 21 would fit you best.”

The follow-up is superficially reasonable but remains shallow. The system does not probe what “comfortable” means here, why removability may be relevant, whether layering is the issue, or what trade-off the user is managing between office appropriateness and ease of wear. Across sessions, the baseline could ask reasonable opening questions, but rarely clarified the underlying decision criteria in much depth.

Stage 2: Recommendation failures

W3: Poor reasoning over recommendation sets. Once recommendation began, the baseline often failed to treat grouped candidates as a coherent set with a shared rationale that could later be refined, narrowed, or discarded. As the VOGUE analysis suggested, multimodal recommendation often proceeds through grouped candidates that are introduced together, critiqued together, and then reduced through comparison. The baseline, however, often treated items too independently once feedback arrived. One example appears in a dinner-party interaction:

Assistant: “I’d put you in either: - item 13 for a relaxed, classic sweater look, or - item 19 if you want that extra ‘wrapped up by the fireplace’ feel with the shawl collar.”

User: “mmm i feel like i dont do so well in anything that hugs the body too tightly”

Assistant: “Got it—let’s keep it cozy but not clingy. For a softer, looser feel, I’d steer you to a regular-fit cardigan like item 19 or item 24 that you can wear slightly open.”

The user’s objection to tight-fitting clothing should have prompted reconsideration of the earlier recommendation frame. Instead, the system returned to item 19 under a new description, even though that item belonged to the same now-questionable set. More broadly, feedback on one member of a grouped recommendation often failed to trigger reconsideration of the shared rationale behind the set. The model therefore treated item mentions too independently rather than maintaining recommendation sets as conversational hypotheses.

Stage 3: Critique and refinement failures

W4: Weak use of implicit and contrastive feedback. After making recommendations, the baseline was much more reliable with explicit statements than with weaker conversational evidence. In realistic recommendation dialogue, users often communicate preferences through contrast, hesitation, selective enthusiasm, or mild rejection rather than clean declarative statements. The baseline used these signals inconsistently. Negative or lukewarm feedback did not always produce durable updates, and items similar to already rejected ones sometimes reappeared under slightly different framing. This compounded the set-level problem above: once the model had committed to a local direction, it often treated critique as a comment on one item rather than as evidence about the broader recommendation logic.

W5: Grounding without strong decision support. Because the catalogue was small and fully visible in context, the system usually stayed close to the available items and did not often hallucinate unsupported properties. But grounding alone did not make the interaction strong. Responses were still repetitive, wordy, and weakly directed, which made it harder to compare options, reduce uncertainty, and converge on a final choice. The main limitation was the ability to use that evidence coherently across turns to clarify INs and narrow towards a final choice.

These observations motivate the more structured systems introduced next. L1 addresses the most immediate continuity failures by externalizing lightweight persistent state over user preferences, item history, and coarse interaction policy. L2 extends this by explicitly representing unresolved INs, dialogue-level observations, and recommendation-set history. More broadly, the progression from the baseline to L1 and L2 reflects a shift from minimally structured response generation toward interaction state that more directly represents what remains unresolved in the decision.

B.2.2 Comparative qualitative analysis across systems

We next compare the baseline, L1, and L2. As in the main chapter, we organize the discussion by the interaction stage where the clearest qualitative differences appeared. Both design variants strengthen contrasting elements of recommendation; L1 mainly strengthens carryover of already surfaced preferences and constraints, whereas L2 more often achieved improves by operationalizing INs after recommendation and critique.

Stage 1: Elicitation

The clearest improvement from the baseline to the structured variants appeared in elicitation. Relative to the unstructured condition, both L1 and L2 were less likely to collapse clarification and recommendation into the same move, and more often used an additional turn to narrow the decision space before introducing candidates.

L1’s main gain came from improved short-horizon continuity. Because it explicitly tracked basic preference and policy state, it more consistently preserved surfaced exclusions and practical constraints across turns. In the city-walking shoe scenario, for example, once the user rejected the “gym shoe” look and specified dislike for cloth uppers and white soles, L1 moved away from that style family and shifted toward leather- or suede-based alternatives rather than drifting back toward the rejected direction. Across sessions, L1 usually produced a more orderly narrowing process when the user’s requirements were concrete and easy to verbalize.

That gain was concentrated in continuity rather than diagnosis. L1 was better at carrying forward surfaced signals than at identifying unresolved INs that had not yet become explicit in the dialogue. Its main contribution at this stage therefore lay in continuity rather than in deeper clarification of what still had to be resolved.

L2 produced the clearest change in elicitation behavior. Compared with L1, its follow-up questions more often targeted unresolved INs. Where L1 mainly preserved surfaced constraints, L2 more often treated hesitation or critique as evidence that a decision-relevant question remained open. That pattern was especially visible in style-heavy cases where the user’s preference could not be reduced to a single attribute. Rather than only narrowing among nearby items, L2 was more likely to ask what key distinction still had to be resolved for the decision, such as

silhouette, texture, layering, formality, or a broader stylistic criterion. The schema changes in L2 more often guided elicitation toward the IN still blocking progress.

Stage 2: Recommendation

Recommendation remained difficult across all three systems. The baseline most often failed by treating proposed items too independently. When two or more candidates were introduced under a shared rationale, later feedback on one item rarely led the system to reconsider that rationale at the level of the set. Recommendation sets therefore behaved less like coherent conversational hypotheses and more like loose clusters of item mentions.

L1 changed that pattern only slightly. It improved local continuity and sometimes narrowed candidate sets more coherently, but it still tended to reason one item at a time. In many sessions, the system replaced a rejected item with a nearby substitute while leaving the broader recommendation frame largely unchanged. That made recommendation somewhat more orderly than in the baseline, yet still left the relevant IN weakly represented at the level of the set. When a grouped recommendation failed, L1 often responded by repairing the last item choice rather than revisiting the shared criterion that had motivated the set. That made recommendation somewhat more orderly than in the baseline, yet still left the relevant IN weakly represented at the level of the set.

L2 showed partial improvement here as well. Because it explicitly tracked shown sets together with richer dialogue observations and unresolved INs, it was better able to treat grouped recommendations as structured moves within an evolving decision process. In practice, that sometimes produced stronger follow-up after a set had failed, especially when user critique helped surface the unresolved IN that the earlier set had missed. Even so, the transcripts still fall short of strong set-level reasoning. In several sessions, the interaction remained largely sequential. Much of the gain came from better clarification and repair around emerging INs, rather than from stable management of recommendation sets as full conversational hypotheses.

Stage 3: Critique and refinement

The strongest differences between the three systems emerged once users began reacting to candidate items. At that point, the baseline was much more reliable with explicit statements than with weaker conversational signals. It often treated critique as a narrow comment on one item rather than as evidence about the recommendation frame more broadly. Softer feedback such as selective enthusiasm, partial rejection, or contrast within a set also carried little weight in later turns. In IN terms, the baseline often failed to treat critique as evidence that an unresolved question had become newly salient.

L1 improved that pattern mainly through better persistence of explicit feedback. Once the user directly ruled something out, L1 was more likely than the baseline to preserve that

exclusion across subsequent turns. In several sessions, that produced cleaner local repair and reduced obvious forgetting. Even so, the system remained largely reactive and item-local. It often handled critique by finding a nearby substitute rather than by asking which IN the failed recommendation had exposed.

L2 again showed the clearest improvement. Compared with L1, its critique handling more often targeted unresolved INs. That shift was clearest in the dinner-party session, where multiple adjacent options were rejected. Instead of continuing to cycle through nearby garments, L2 reformulated the problem around the missing stylistic criterion and asked whether the user wanted more texture, pattern, a different neckline, or a different cut. A similar pattern appeared in the city-trip shoe session. L2 initially over-indexed on weather and suggested heavier, boot-like options, but once the user rejected that framing as overkill, it reopened the candidate space and shifted toward a more sneaker-like subset. In each case, the main value of L2 lay in interpreted critique. It more readily treated critique as evidence that the earlier framing, and then used that diagnosis to redirect the interaction.

B.2.3 Summary of qualitative gains and remaining limits

Taken together, the strongest gains from adding structured state appeared in continuity, elicitation, and critique handling. L1’s main contribution was better short-horizon continuity: once practical constraints, exclusions, or explicit preferences had been surfaced, it was more likely to preserve them across turns. L2 extended this in a more decision-relevant direction. By more often treating hesitation, contrast, or critique as evidence that an important criterion via INs, it produced stronger follow-up, more selective refinement, and better recovery when the initial recommendation frame proved incomplete.

At the same time, recommendation-set reasoning remained difficult across all three systems. The baseline often treated grouped candidates too independently, and L1 did not substantially change this. L2 showed partial improvement, especially when INs helped surface reasoning gaps, but recommendation sets still did not behave consistently as structured conversational hypotheses. More broadly, the structured additions did not fully solve the recommendation problem: even where L2 improved critique handling and remained close to L1 on the alignment metrics, these gains did not translate into uniformly strong comparison, narrowing, or final closure. Together, these qualitative observations support the interpretation in the main body that richer explicit state helps, and that representing unresolved INs can materially improve recommendation behavior, while still falling short of strong end-to-end decision support.